

Air cargo scheduling: integrated models and solution procedures

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Published online: 15 June 2012
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Abstract Designing a profitable flight schedule is a highly complex planning problem. Both passenger and cargo airlines usually follow a decomposition approach and break this problem into several subproblems which are then solved consecutively and iteratively using specific but isolated models. At cargo airlines, the four major interdependent decision problems are flight selection, fleet assignment, rotation planning, and cargo routing. In our research, we have developed a planning approach which differs from other OR-based planning approaches in two aspects. The approach is based on integrated models and it is based on the pragmatic planning paradigm to optimally modify an existing schedule. For this purpose, the planner has to identify mandatory and optional flights. Then the planning goal is to identify the best combination of optional flights to be included into the schedule. Our integrated planning models comprise several additional important planning aspects for cargo airlines such as available capacities on external flights (e.g. belly capacities from passenger flights or road-feeder services), cargo handling costs and constraints, and aircraft maintenance regulations. There are two main aspects which we present in this paper. First, we describe the planning problem and the specific planning paradigm, develop a set of complex mixed-integer programs representing the different subproblems, and finally present integrated problem formulations as well as several model extensions. Thereafter, we develop a branch and price and cut approach for solving the mathematical programs and present extensive computational results obtained for a set of generated

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yet highly practical problem instances for different types of carriers. The results show that our approach is able to find high quality solutions to problem instances of realistic size and complexity within reasonable time.

Keywords Air cargo transportation · Schedule planning · Fleet assignment · Rotation planning · Cargo routing · Branch and price and cut

1 Introduction

A cargo airline offers the conceptually simple service to transport goods from an origin to a destination within a certain time interval at a given price. For this purpose, the airline keeps a fleet of aircrafts and operates a set of flights which are published in the flight schedule. Note that there are subtle but significant differences between the passenger and the cargo business: while passengers usually book return tickets, a cargo booking is one way which possibly leads to imbalanced markets; while passengers book complete itineraries, i.e. sequences of legs from an origin to a destination, a cargo booking is generally just an origin–destination pair with a time constraint and hence the airline has some freedom to route goods from their origin to their destination. Thus, market demand can be described as a set of *O&Ds* or *O&D pairs*.

A flight schedule determines the transportation potential of the airline as well as its resource requirements and consumption and thus it is the central planning entity from both a market oriented point of view and a production point of view. It describes the transportation network, i.e. which airports are connected by a direct flight (called *leg*) and which *O&D pairs* can be connected by a path of legs, as well as the traffic on the network, i.e. when (timing) and how often (frequency) the legs are operated during a week. It determines which aircraft type operates a certain leg at a certain time as well as the production process, i.e. which sequences of legs are operated by the same airplane to enable a periodic schedule over the weeks of the planning period. Once published, yield potential and operating costs are predetermined since a flight schedule can only be altered marginally and/or at high cost. Designing a profitable and feasible schedule is extremely complex in scope and size since it affects decisions on all resources and thus “typically involves the whole airline”. Therefore, [Etschmaier and Mathaisel \(1985\)](#) have propagated the concept of an iterative planning process based on two phases: schedule construction and schedule evaluation.

Due to the complexity of constructing a feasible schedule, airlines typically follow a decomposition approach to break up the schedule planning problem into several smaller problems which are solved sequentially. The majority of literature on passenger airline schedule planning identifies at least four sequential steps: *schedule design*, *fleet assignment*, *aircraft rotation planning/aircraft routing*, and *crew scheduling* (see e.g. [Gopalan and Talluri 1998](#); [Barnhart and Cohn 2004](#); [Gopalakrishnan and Johnson 2005](#)). In general, cargo airlines follow a similar planning process (see [Antes 1998](#); [Zils 2000](#)) which is illustrated in Fig. 1. However, evaluating a cargo airline schedule is rather complex compared to evaluating a schedule for passenger airlines. As cargo has no preference for a particular itinerary or route, an additional problem, the *cargo routing problem*, must be solved to identify optimal cargo itineraries.

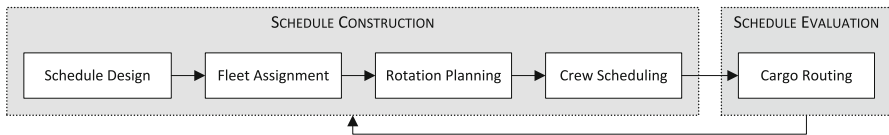


Fig. 1 The air cargo schedule planning process

While air scheduling problems for passenger airlines have been discussed in the OR literature extensively since several decades, contributions which specifically focus on the air cargo industry are still relatively rare. [Gopalan and Talluri \(1998\)](#) and [Barnhart et al. \(2003\)](#) give an overview of the general flight schedule planning process and the various subproblems arising in passenger airlines. A comprehensive exposition of the planning process and the problems arising in air cargo planning is given by [Zils \(2000\)](#) who also proposes a complex decision support framework for integrating the entire planning process and supporting different decisions at the various stages. [Yan et al. \(2006\)](#) discuss a model for airport selection and timetable setting. [Marsten and Muller \(1980\)](#) introduce a mixed-integer model for air cargo fleet assignment, while [Antes et al. \(1998\)](#) focus on the aspect of cargo routing. [Li et al. \(2006\)](#) present a model for a fleet assignment problem faced by a major Asia-Pacific mixed carrier with six different passenger fleets and a single cargo fleet. A model for combined passenger and cargo flight scheduling is presented by [Tang et al. \(2008\)](#).

Our approach to air cargo scheduling differs from most planning approaches presented in the Operations Research literature with respect to two paradigms: first, we do not plan on the level of single flight legs but on the level of predefined flights, i.e. sequences of flight legs, and second, we assume that the planning goal is to refine and adapt an existing schedule instead of optimizing from scratch. We call this approach pragmatic since it respects the combinatorial complexity of the problem and reflects the planning practices at cargo airlines. [Lohatepanont and Barnhart \(2004\)](#) present a similar approach for passenger airline schedule planning.

The objective of our research is to give a proof of concept of the pragmatic planning approach by developing integrated models and state-of-the-art solution methods. The approach has been developed primarily to support planning at typical cargo airlines, such as Lufthansa Cargo, Cargolux, or Singapore Airlines Cargo. These airlines operate around 10–30 aircrafts to serve a worldwide hub and spoke network consisting of several 100 legs. However, our models and methods can equally be applied to smaller regional carriers as well as express carriers such as DHL or Federal Express which operate more than 100 aircrafts on a few 1,000 legs. The complexity of the specific network of a cargo airline, i.e. the number of destinations, flights, and airplanes as well as the heterogeneity of the fleet, determines the complexity and size of typical problem instances.

The computational tractability and effectiveness of our approach is shown through extensive tests on realistic data representing different types of cargo airlines. Finally, a fully functional decision support system based on the models and methods is developed allowing user-friendly analysis of various scenarios in strategic as well as tactical planning. An in-depth description of the air cargo schedule planning problem, integrated models and solutions procedures as well as computational results can be found in [Friederichs \(2010\)](#).

Our paper is partitioned as follows. First, we present our pragmatic planning approach and a conceptual description of the specific and complex scheduling problem arising at cargo airlines in Sect. 2. Then, we develop a set of models representing different subproblems as well as an integrated model in Sect. 3, followed by the introduction of several model extensions which are of practical relevance in Sect. 4. Then we focus on the solution procedures which we have developed to solve the integrated models. After introducing the basic branch and price and cut procedure in Sect. 5, we describe the most important features of our implementation with column generation in Sect. 6, branching strategies in Sect. 7 and cutting planes in Sect. 8. Computational results obtained on a rich set of problem instances are presented in Sect. 9. In Sect. 10, we give some final remarks.

2 The air cargo scheduling problem (ACSP)

In this section, we first motivate the pragmatic planning approach and illustrate how it is implemented and we define the Air Cargo Scheduling Problem (ACSP) on a conceptual level. Then, in the following section, the ACSP is defined in greater detail.

2.1 The pragmatic planning paradigm

In our approach, we leave the generation/adaption of flights in the hand of the experienced planners and support the schedule design process by a model-based system which selects the most profitable combination of a set of predefined flights. The motivation for the decision to plan on the level of predefined flights instead of single legs is twofold. First, the network and demand structure of cargo airlines makes it natural to think in terms of flights. In contrast to passenger airlines, the (aggregated) demand from/to a single airport is usually rather small compared to the capacity of cargo aircrafts. Consequently, cargo aircrafts often start at a hub, visit a sequence of destinations at which cargo is collected and/or delivered, and then arrive at a (potentially different) hub at which cargo may be transferred. In the context of this paper, we define such a sequence of legs departing from and arriving at a hub as a *flight*. Second, planning based on legs may certainly be adequate in cases where the planning goal is to generate a completely new network structure. Yet, since such changes in an airline's network structure require significant investments at airports, organizational effort, and the availability of information airlines do not want to change their network and schedule significantly from one period to the next. Even more important, customers value a continuity in services highly since it enables them to plan their shippings in advance. This is especially true for the business markets where reliable services are mandatory, e.g. in supply chains with just-in-time delivery. To maintain such a continuity in operations and services, the most recent schedule is used as a basis and modifications are introduced to this schedule to adapt to a changing environment. In our approach, we leave the generation/adaption of flights in the hand of the experienced planners and support the schedule design process by a model-based system which selects the most profitable combination of a set of predefined flights.

Thus, we differentiate between mandatory and optional cargo flights. *Mandatory* cargo flights are fixed, i.e. they have to be part of the final schedule whereas *optional* cargo flights should only be included if they produce an additional contribution to profit. In addition to mandatory and optional cargo flights, a cargo airline may have access to *external* flights, i.e. flights which stem from an airline's passenger fleet, partner airlines, or road-feeder services. Combined, mandatory, optional, and external flights form a *master flight list*. The process of constructing a master flight list starts with the list of all cargo flights operated in the current schedule. All core flights which constantly yield a high return and/or load factor should be part of the mandatory flight list, while flights which did not yield the expected return or flights for which there exist alternative flights are candidate flights for deletion. Then a list of potential new cargo flights is generated and put into a list with candidate flights for addition. This list is the central means for planners to introduce changes to the current schedule. Here, flights which add services to new airports or markets can be included as well as flights which form alternatives to current flights (which should be marked as candidates for deletion). As mentioned above, external flights are flights which are available for cargo transportation but are not conducted by the airline's cargo fleet. Hence, they do not need to be considered in fleet assignment and rotation planning. Now the master flight list is constructed by merging mandatory cargo flights, optional cargo flights, and a list of external flights. Given this master flight list, the basic problem is to identify the combination of optional cargo flights which complement the mandatory cargo and external flights such that by satisfying (part of) the expected demand contribution to profit is maximized. Cargo airlines define markets and market demand in form of O&D's where a single O&D denotes the transportation demand from an origin airport to a destination airport. Note that in the planning phase demand plays the role of a projected market demand which functions as an upper bound and which does not have to be met completely. Thus, the air cargo scheduling problem (ACSP) requires to

- Select a combination of optional cargo flights (schedule design),
- Assign each mandatory and selected optional flight to a fleet type (fleet assignment),
- Create feasible rotations for the aircrafts of each fleet type (aircraft rotation), and
- Identify which O&D's to fulfill over which itineraries (cargo routing).

Fleet assignment and aircraft rotation planning must be considered to guarantee the feasibility of the resulting schedule and to allocate capacities according to the demand. The cargo routing problem is an essential part of the entire planning process since the demand and its profitable fulfillment over feasible routings drive the selection of optional flights. Obviously iterative planning with isolated models for each sub-problem becomes impractical since all models had to be solved for every combination of optional flights, i.e. $2^{|N|}$ times, where N is the set of optional flights. Note that the crew scheduling problem is not taken into account since cargo airplanes are usually operated by two pilots without any cabin crew. Compared to crew scheduling for passenger flights with up to 15 flight attendants, cargo crew scheduling is not a crucial aspect and should not influence a solution of the ACSP. The following subsections define the ACSP planning problem in detail.

2.2 Conceptual description of ASCP

In this section, we specify the ASCP describing its input and output, the planning objective, the assumptions it is based upon, as well as the constraints a feasible solution must comply with.

2.2.1 Problem data

The following list specifies the required model input data:

- The set of airports with associated attribute values such as the minimal ground time for each fleet type and the minimal transfer time and the storage costs of cargo.
- The master flight list with fixed and variable transportation costs for each flight.
- The fleet types along with their maximal capacity in terms of weight and volume and the number of available airplanes.
- The O&D potential of the airline given by O&D requests with information on the demand weight and volume, the freight rate, as well as its earliest pickup and its maximal delivery time.

The main output consists of the final flight schedule which includes all selected cargo flights, feasible rotations for each fleet type that cover all flights in the schedule, and the cargo routings, i.e. the cargo flow for all demands from their origin to their destination. Given a solution, the profit/loss calculation and other relevant information such as the average payload on flights or the yield per unit shipped can be calculated easily.

2.2.2 Assumptions

The ACSP is taking into account the several assumptions which are in line with common practices. The set of flights in the master flight list as well as the demand pattern repeats every T time units, i.e. the resulting schedule is periodic where most long-haul cargo airlines construct a weekly schedule. The *flight time* is independent of the actual aircraft type assigned to the flight. The demand of an O&D is arbitrarily divisible and can be shipped via several O&D paths. Fuel costs on a leg can be divided into a *fixed part* i.e. the fuel costs incurred when conducting a flight without any payload depending on the fleet type and a *variable part* which increases proportionally with the actual payload. Note that although variable fuel costs depend on the fleet type, we assume this part to be independent of the assigned fleet. However, to compensate for this, fleet-dependent fixed fuel costs for each flight may be raised (lowered) to cover for significantly higher (lower) fuel burn rates. For models that do not enforce cargo handling constraints explicitly, the time needed to load or unload cargo from an aircraft is assumed to be shorter than the aircraft ground time at the airport. Cargo transfer costs increase proportionally with the amount of cargo transferred depending on the O&D category and the airport the cargo is transferred at. Cargo storage costs increase proportionally with the time and amount of cargo stored depending on the O&D category and the airport the cargo is stored at. The storage capacity at an airport

is not taken into account and is assumed to be unlimited. All aircrafts of the same fleet type are assumed to be equal in terms of fixed costs, maintenance costs, and maintenance requirements. Costs already paid like leasing costs for leased airplanes etc. are either decision irrelevant or can be prorated and are assigned to one of the fixed or variable cost components.

2.2.3 Planning objective

While fleet assignment and rotation planning aim at minimizing the costs of a schedule, the goal of cargo routing is to maximize the revenue generated. Thus, the overall planning objective of the ACSP is to maximize the projected net profit generated with the proposed schedule. Here, *revenue* is calculated by summing up the freight rate of all O&D requests multiplied by the actual amount transported. The cost term is composed of three components. *Variable transportation costs* include costs which depend on the payload on a flight as well as cargo handling and storage costs incurred at airports. *Fixed flight costs* are the payload-independent costs of flights if conducted by a certain fleet type. These costs include the landing costs at airports, en-route-fees as well as the part of the fuel costs incurred if the flight is conducted with no freight loaded. *Fixed fleet costs* are costs which can be accounted to a single aircraft of a given fleet type, e.g. leasing fees or the initial investment costs broken down to the planning horizon.

2.2.4 Constraints

A solution of the ACSP is subject to various constraints of different complexity. There are several types of constraints which we consider common to all cargo airlines, while other constraints may only pertain to certain airlines or planning scenarios. Also, the requirements of a specific planning situation affect the choice of which constraints to enforce. Therefore, we differentiate between *basic constraints* which every schedule must comply with and *optional constraints* which can be enforced if needed. In the following, we list all constraints according to the subproblem they arise in. With every constraint we associate an abbreviation which is used as reference in the mathematical models.

In *fleet assignment*, we have to obey the following constraints. Each mandatory and selected optional cargo flight must be assigned to exactly one fleet (FA-1-COVER). The number of available aircrafts of each fleet must not be exceeded (FA-2-SIZE). A flight can only be conducted by a compatible fleet type. There are several reasons for incompatibilities, e.g. if the maximal range of the aircraft is shorter than the range of any of the legs belonging to the flight or the runway of any of the visited airports is shorter than the required runway length for the fleet type (FA-3-COMP). The number of times an airport is served within the planning horizon must be at least as high as the minimal frequency and at most as high as the maximal frequency specified for the airport (FA-4-FREQ)

A *rotation* has to fulfill the following conditions. Each airplane must follow a cycle/rotation of flights which is repeated periodically (RP-1-CYCLE). The sequence of flights in a rotation must be connected, i.e. the arrival airport of a flight must be the departure airport of the following flight (RP-2-CONN). The ground time at an

airport between two consecutive flights in a rotation must be at least as long as the minimal ground time of the conducting fleet type at the airport (RP-3-GROUND). A maintenance stop is necessary as soon as the maximal number of flight hours, the maximal number of cycles (take-offs and landings) or the maximal elapsed time since the last maintenance stop is reached (RP-4-MNT). Maintenance can only be performed at airports which offer maintenance services for the fleet type (RP-5-MNTHUB). To enforce an equal wear-and-tear, all aircrafts of a fleet must follow a single rotation (RP-6-EQUAL).

Finally, *cargo routing* is subject to the following constraints. The total amount shipped over the cargo flow paths of an O&D must not exceed the demand volume in terms of weight (CR-1-DEMAND). The payload on a flight leg in terms of weight and volume must not exceed the maximal weight and maximal volume of the fleet type conducting the flight (CR-2-PAYLOAD). A cargo flow path must be a sequence of connected flight legs in which each arrival airport of a leg is the departure airport of the following leg (CR-3-CONN). An O&D can only be shipped from its origin when its earliest pickup time is reached (CR-4-AVAILTIME). An O&D must arrive at its destination before its latest delivery time (CR-5-DELTIME). Cargo can only be transferred and stored at airports with adequate handling/storage facilities for the O&D (CR-6-XFRHUB). Cargo can only be transferred between two flights if the time between the arrival of the first flight and the departure of the subsequent flight is at least as long as the minimal transfer time of the O&D at the airport (CR-7-XFRTIME).

3 Mathematical models

3.1 Basic concepts

3.1.1 Planning horizon

The planning horizon for a schedule is given by the sequence $T = \{1, 2, 3, \dots, N\}$ of points in time reflecting the duration of a periodic plan. Thus, for a weekly schedule planned to-the-minute, $N = 7(\text{days}) \times 24(\text{h/day}) \times 60(\text{min/h}) = 10,080$. Since the created plan is periodic, we introduce connections from a point in time $t_1 \in T$ to an earlier point in time $t_2 \in T, t_2 < t_1$ representing connections spanning over two weeks, i.e. we interpret this connection as if t_2 is the corresponding point in the following period. Thus, we define a function $\text{time}(t_1, t_2)$ which calculates the time from t_1 to t_2 as follows:

$$\text{time}(t_1, t_2) = \begin{cases} t_2 - t_1 & \text{if } t_2 \geq t_1 \\ t_2 - t_1 + N & \text{otherwise} \end{cases}$$

3.1.2 Airports

Let A be the set of all airports or destinations served by any cargo flight or external flight. Then, we set a binary parameter $\text{hub}_a = 1$ if cargo may be transferred at airport $a \in A$ and $\text{hub}_a = 0$ else. In the following, we refer to an airport which allows cargo to be transferred as a *transfer hub*.

3.1.3 Fleet types

Let K be the set of fleet types with $n_k \in \mathbb{N}$ the number of available airplanes of type k , $c_k^{\text{acr}} \in \mathbb{R}^+$ the fixed costs and $w_k^{\text{max}} \in \mathbb{R}^+$ and $v_k^{\text{max}} \in \mathbb{R}^+$ the maximal payload in terms of weight (tonnes) and volume (m^3), respectively, for each aircraft of fleet $k \in K$. Once landed at an airport, each aircraft must stay on ground for some time to allow for activities such as loading and unloading of cargo, fueling, or basic security checks. This *minimal ground time* depends on the fleet type k as well as the airport a and is given by $\text{gt}_{a,k}^{\text{min}} \in \mathbb{N}$.

3.1.4 Legs

A *flight leg* l is a direct connection from an origin airport $\text{orig}_l \in A$ to a destination airport $\text{dest}_l \in A$ and the departure and arrival times are given by $\text{dep}_l \in T$ and $\text{arr}_l \in T$, respectively. Note that the departure/arrival time is the moment the airplane leaves/arrives at its parking position and the duration between the departure and the arrival time is the *block time* $\text{dur}_l := \text{time}(\text{dep}_l, \text{arr}_l)$. The flight time $\text{ft}_l \in \mathbb{N}$ of a leg is defined as the block time minus the taxi times. Now the set L of all legs is partitioned into two disjoint sets L^E and L^C , with L^E being the set of all external flight legs and L^C being the set of all cargo flight legs. While for cargo flight legs $l \in L^C$, the maximal payload weight $w_l^{\text{max}} \in \mathbb{R}^+$ and the maximal payload volume $v_l^{\text{max}} \in \mathbb{R}^+$ depend on the fleet type the flight is assigned to, these capacities are known in advance for all external flight legs $l \in L^E$. As argued in Sect. 2.2.2, we assume that for a flight leg $l \in L$ the variable transportation costs $c_l^{\text{leg}} \in \mathbb{R}^+$ given in US\$ per tonne are independent of the fleet type while the fixed costs are part of the fixed costs of the flight the leg belongs to.

3.1.5 Flights

We define a *flight* or *route* f as a sequence of legs $(l_1^f, l_2^f, \dots, l_{n_f}^f) \in L^{n_f}$, with $n_f \in \mathbb{N}$ being the number of legs of f . Given a leg l , f_l denotes the flight l is part of. Let F be the set of all flights, then we differentiate between the set of external flights F^E which are available for cargo routing and the set of cargo flights F^C which must be considered in fleet assignment and aircraft routing. Each external flight $f \in F^E$ consists of external legs $l_i^f \in L^E, i = 1, \dots, n_f$, while each cargo flight $f \in F^C$ consists of cargo flight legs $l_i^f \in L^C, i = 1, \dots, n_f$. While we assume that each external flight $f \in F^E$ is feasible, we have to secure for all cargo flights $f \in F^C$ that the necessary condition that the arrival airport of a leg must be the departure airport of the following leg is fulfilled. A feasible cargo flight f may be incompatible with a fleet k , e.g. if the time between two consecutive legs is shorter than the minimal ground time of the fleet k at the airport or the longest leg of f exceeds the maximal range of the fleet k . Thus, we define binary parameters $\text{comp}_{k,f} \in \{0, 1\}$ which take the value 1 if f can be assigned to fleet k , 0 otherwise. Then, $F_k^C := \{f \in F^C | \text{comp}_{k,f} = 1\}$ is the set of cargo flights compatible with fleet k . Moreover, since we differentiate between mandatory and optional cargo flights, we introduce binary parameters mdt_f

which indicate if a cargo flight f is mandatory ($=1$) or optional ($=0$). The fixed costs of a cargo flight $f \in F^C$ depend on the aircraft type k the flight is assigned to and are given by $c_{k,f}^{\text{flt}} \in \mathbb{R}^+$. The origin airport orig_f and the departure time dep_f of a flight $f \in F$ are the origin airport and the departure time of the first leg of the flight while the destination airport dest_f and the arrival time arr_f of f are given by the destination airport and arrival time of the last leg of f , respectively. Finally, the total duration of a flight is given by $\text{dur}_f := \text{time}(\text{dep}_f, \text{arr}_f)$, and the flight time ft_f is the sum of the flight times on all legs of f .

3.1.6 Rotations

We define a rotation r as a sequence of connected flights operated by a fleet $k_r \in K$, i.e. as a tuple $(k_r, f_1^r, f_2^r, \dots, f_{n_r}^r) \in K \times F^{n_r}$, where $n_r \in \mathbb{N}$ is the number of flights in r . Let R be the set of all rotations. We say a rotation r is feasible if

- Each flight is compatible with k_r , i.e. $\text{comp}_{f_i^r, k_r} = 1, \forall i = 1, \dots, n_r$,
- The flights are connected, i.e. $\text{dest}_{f_i^r} = \text{orig}_{f_{i+1}^r}, \forall i = 1, \dots, n_r - 1$,
- The rotation is cyclic which implies that the arrival airport of the last flight must be the departure airport of the first flight, i.e. $\text{dest}_{f_{n_r}^r} = \text{orig}_{f_1^r}$, and
- The minimal ground time at all airports between the flights is respected, i.e. $\text{time}(\text{arr}_{f_i^r}, \text{dep}_{f_{i+1}^r}) \geq \text{gt}_{k_r, \text{dest}_{f_i^r}}^{\min}, \forall i = 1, \dots, n_r - 1$ and $\text{time}(\text{arr}_{f_{n_r}^r}, \text{dep}_{f_1^r}) \geq \text{gt}_{k_r, \text{dest}_{f_{n_r}^r}}^{\min}$.

3.1.7 ODs

Let OD be the set of estimated or booked service requests which define the market demand, commonly referred to as *O&Ds* or *O&D pairs*. Then, each O&D pair $\text{od} \in \text{OD}$ is associated with an origin and a destination airport, denoted by $\text{orig}_{\text{od}} \in A$ and $\text{dest}_{\text{od}} \in A$, respectively. Additionally we associate two dates, $\text{avail}_{\text{od}} \in T$ is the time an O&D $\text{od} \in \text{OD}$ is available at its origin and $\text{dt}_{\text{od}}^{\max} \in \mathbb{N}$ is the latest allowable delivery time of an O&D od at its destination. The demand weight $w_{\text{od}} \in \mathbb{R}^+$ of od is measured in tonnes and the demand volume $v_{\text{od}} \in \mathbb{R}^+$ is measured in cubic meters. The yield or freight rate $r_{\text{od}} \in \mathbb{R}^+$ is given in US\$ per tonne.

3.1.8 Cargo flow paths

We define a *cargo flow path* or *itinerary* p as a sequence of connected legs which may be used to ship an O&D $\text{od} \in \text{OD}$, i.e. as a tuple $(\text{od}_p, l_1^p, l_2^p, \dots, l_{n_p}^p) \in \text{OD} \times L^{n_p}$, where $n_p \in \mathbb{N}$ is the number of legs in p . We say a path $p \in P$ is feasible if

- The origin airport of od_p is the departure airport of the first leg, i.e. $\text{orig}_{\text{od}_p} = \text{orig}_{l_1^p}$,
- The destination airport of od_p is the arrival airport of the last leg, i.e. $\text{dest}_{\text{od}_p} = \text{dest}_{l_{n_p}^p}$,
- The legs are connected, i.e. $\text{dest}_{l_i^p} = \text{orig}_{l_{i+1}^p}, \forall i = 1, \dots, n_p - 1$,

- Cargo is only transferred at transfer hubs, i.e. $f_{l_i^p} \neq f_{l_{i+1}^p} \Rightarrow \text{hub}_{\text{dest}_{l_i^p}} = 1, \forall i = 1, \dots, n_p - 1$, and,
- The maximal delivery time is not exceeded, i.e. $\text{time}(\text{avail}_{\text{od}}, \text{dep}_{l_1^p}) + \sum_{i=1}^{n_p-1} \left(\text{dur}_{l_i^p} + \text{time}(\text{arr}_{l_i^p}, \text{dep}_{l_{i+1}^p}) \right) + \text{dur}_{l_{n_p}^p} \leq \text{dt}_{\text{od}_p}^{\max}$.

Note that stating the departure and arrival airport of a path is not sufficient to identify the corresponding od since there might be several O&Ds with varying rates, availabilities and/or time constraints for these airports. Given the set of feasible cargo flow paths P , $P_{\text{od}} := \{p \in P \mid \text{od}_p = \text{od}\}$ is the set of paths for the O&D od. In the basic models, we assume that flights are timed so that there is always enough time to transfer cargo between two flights even if they are not operated by the same aircraft. Finally, we define the margin $m_p \in \mathbb{R}$ on a path p as the rate of the corresponding od minus the variable transportation costs on the path, i.e. $m_p := r_{\text{od}_p} - \sum_{l \in L_p} c_l^{\text{var}}$.

3.1.9 Flight strings

To model maintenance constraints, we apply the concept of flight strings which was introduced by Barnhart et al (1998). A *flight string* or *string* is a sequence of connected flights which starts and ends at maintenance hubs and is maintenance feasible, i.e. the flights of the string can be conducted without any maintenance check in between. An *augmented flight string* is a flight string with the minimum time necessary to perform maintenance attached to the end of the last flight in the string. Unless stated otherwise, we say flight string instead of augmented flight string in the following.

We define a flight string $s \in S$ as a tuple $(k_s, f_1^s, f_2^s, \dots, f_{n_s}^s) \in K \times F^{n_s}$ with $n_s \in \mathbb{N}$ the number of flights and $k_s \in K$ the conducting fleet type. Let $\text{orig}_s \in A$ be the origin airport of the first flight in s and dest_s be the destination airport of the last flight in s . Then, a flight string s is feasible if

- Each flight is compatible with k_s ($\text{comp}_{k_s, f_i^s} = 1, \forall i = 1, \dots, n_s$),
- The flights are connected, i.e. $\text{dest}_{f_i^s} = \text{orig}_{f_{i+1}^s}, \forall i = 1, \dots, n_s - 1$,
- The origin airport of the first flight is a maintenance airport for the fleet type k_s ($\text{hub}_{k_s, \text{orig}_s}^{\text{mnt}} = 1$) and the destination airport of the last flight is a maintenance station for the fleet type k_s ($\text{hub}_{k_s, \text{dest}_s}^{\text{mnt}} = 1$),
- The minimal ground time at all airports between the flights is respected, i.e. $\text{time}(\text{arr}_{f_i^s}, \text{dep}_{f_{i+1}^s}) \geq \text{gt}_{k_s, a_{f_i^s}}^{\min}, \forall i = 1, \dots, n_s - 1$, and
- The maintenance constraints for the fleet k_s are respected, i.e. if
 - $\sum_{i=1}^{n_s} n_{f_i^s} \leq \text{mnt}_k^{\text{cycles}}$ (maximal number of cycles),
 - $\sum_{i=1}^{n_s} \text{ft}_{f_i^s} \leq \text{mnt}_k^{\text{flight}}$ (maximal flight time), and
 - $\sum_{i=1}^{n_s-1} (\text{dur}_{f_i^s} + \text{time}(\text{arr}_{f_i^s}, \text{dep}_{f_{i+1}^s})) + \text{dur}_{f_{n_s}^s} \leq \text{mnt}_k^{\text{elapsed}}$ (maximal elapsed time).

The fixed costs c_s^{str} of s are the fixed costs of each flight in s plus the costs of the maintenance stop for fleet k at the destination airport of the last flight, i.e. $c_s^{\text{str}} := \sum_{i=1}^{n_s} c_{k_s, f_i^s}^{\text{flt}} + c_{k_s, \text{dest}_s}^{\text{mnt}}$.

Now, a sequence of connected flight strings which begins and ends at the same location is a rotation. Hence, given the set S of feasible flight strings, the problem of finding optimal maintenance-feasible rotations converts into the problem of selecting an optimal set of maintenance-feasible flight strings which can be connected to rotations.

3.2 Modeling the fleet assignment and rotation planning problem (FARP)

There are two well studied approaches for modeling a flight network for fleet assignment and rotation planning (FARP): the connection network model and the timeline network model. Friedrichs (2010) has developed integrated models for solving the ASCP based on both graphical representations for FARP. In the following, we present a flight string model for the FARP with maintenance constraints (FARP-M) based on a timeline network.

First, we introduce the basic (flight) timeline network which is based on *flight arcs* and *ground arcs*. An excerpt of an exemplary timeline network is shown in Fig. 2. Nodes have two dimensions and represent a departure/arrival event of a flight at a specific airport at a specific time. As we model the fleet assignment and aircraft rotation problem on the level of flights, we only consider the arrival/departure airports of flights in the network, while airports which are only serviced on intermediate stops on flights do not need to be considered. Flight arcs connecting two event nodes represent flights assigned to fleet types, i.e. there is one flight arc for each compatible combination of flight f and fleet k , i.e. if $\text{comp}_{k,f} = 1$. Ground arcs are introduced between any two consecutive arrival/departure events for a fleet k at an airport a . Between two such events, the number of airplanes of the fleet on ground at the airport does not change. Let $e_{k,f}^d$ be the departure event of a cargo flight $f \in F^C$ conducted by fleet type $k \in K$

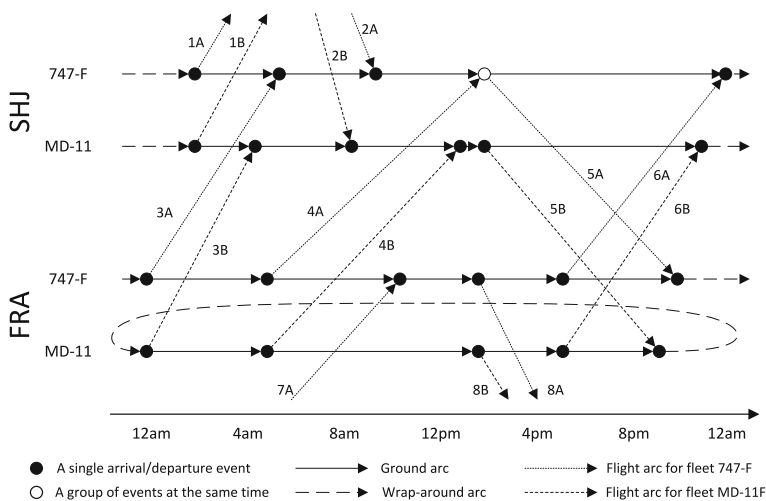


Fig. 2 A timeline network for two airports and two fleets

and $e_{k,f}^a$ the corresponding arrival event. Then, the unique predecessor and successor events connected to the departure (arrival) event $e_{k,f}^d$ ($e_{k,f}^a$) by ground arcs are denoted by $e_{k,f}^{d,-}$ ($e_{k,f}^{a,-}$) and $e_{k,f}^{d,+}$ ($e_{k,f}^{a,+}$), respectively. We define E^G as the set of all ground arcs, and E_k^G as the set of all ground arcs for fleet $k \in K$. If several departure and arrival events of a fleet at a single station occur at the same time, these events are connected using ground arcs (variables) as well, placing arrival events before departure events to allow for connections from arriving to departing flights. Note that the order among arrival (departure) events with the same event time does not matter.

The *string timeline network* contains the same set of nodes i.e. departure and arrival events for each flight and compatible fleet because each flight may be the first or last flight of a flight string. The only difference is that the time of an arrival event is set to the arrival time of the flight plus the minimal maintenance time (instead of the arrival time plus the minimal ground time). Hence, the set of ground arcs E^G is equal. Yet, the model does not contain arcs for flights but arcs for flight strings. Each flight string s is connected with the departure (arrival) event of the first (last) flight of s and fleet k_s . Thus, the departure (arrival) event of a flight $f \in F^C$ and fleet type $k \in K$ is not connected with a single flight arc but with all flight strings for fleet k whose first (last) flight is f . Clearly, all intermediate flights f of a flight string s , i.e. all flights after the first flight and before the last flight of s , are not connected to any arrival/departure events.

For now, assume that all feasible flight strings S are given, i.e. generated a priori. Then the timeline based flight string model *FARP-TS* contains two types of variables. Binary decision variables $y_s^{\text{str}} \in \{0, 1\}$ which take the value 1 if flight string s is selected, and 0 otherwise. Ground variables $y_{e_{k,f}^d, e_{k,f}^d}^{\text{grd}}$ and $y_{e_{k,f}^d, e_{k,f}^a}^{\text{grd}}$ ($y_{e_{k,f}^a, e_{k,f}^a}^{\text{grd}}$ and $y_{e_{k,f}^a, e_{k,f}^d}^{\text{grd}}$) count the number of aircrafts of fleet type k on ground at the departure (arrival) station between the predecessor event and the departure (arrival) event of f and the departure (arrival) event of f and its successor event, respectively. To control fleet size limits, we select a single point in time, the *count time*, at which the number of aircrafts on ground and in the air is counted. The count time can be chosen arbitrarily, but from a computational point of view it is preferred to choose a point in time where the number of aircrafts in the air is minimal. Then, $\chi_s^{\text{str}} \in \mathbb{N}$ is defined as the number of times string s crosses the count time. Note that as rotations may span several weeks, χ_s^{str} may take values higher than 1. Equally, χ_e^{grd} is the number of times ground arc $e \in E^G$ crosses the count time. Now, let S_f be the set of flight strings which cover flight $f \in F^C$, i.e.

$$S_f := \left\{ s \in S \mid \exists i \in \{1, \dots, n_s\} : f_i^s = f \right\}.$$

Additionally, let $S_{k,f}^+ := \{s \in S \mid k_s = k \wedge f_1^s = f\}$ be the set of flight strings for fleet k which start with flight f , and $S_{k,f}^- := \{s \in S \mid k_s = k \wedge f_{n_s}^s = f\}$ be the set of flight strings for fleet k which end with flight f . Using this notation, model *FARP-TS* is defined as follows.

Model FARP-TS

$$\begin{aligned} \min \quad & \sum_{s \in S} c_s^{\text{str}} y_s^{\text{str}} + \sum_{k \in K} \sum_{s \in S} c_k^{\text{acr}} \chi_s^{\text{str}} y_s^{\text{str}} \\ & + \sum_{k \in K} \sum_{e \in E_k^G} c_k^{\text{acr}} \chi_e^{\text{grd}} y_e^{\text{grd}} \end{aligned} \quad (1)$$

$$\text{s.t.} \quad \sum_{s \in S_f} y_s^{\text{str}} = 1 \quad \forall f \in F^C \quad (2)$$

$$\sum_{s \in S_{k,f}^+} y_s^{\text{str}} - y_{e_{k,f}^{d,-}, e_{k,f}^{d,f}}^{\text{grd}} + y_{e_{k,f}^{d,f}, e_{k,f}^{d,+}}^{\text{grd}} = 0 \quad \forall k \in K, \forall f \in F_k^C \quad (3)$$

$$- \sum_{s \in S_{k,f}^-} y_s^{\text{str}} - y_{e_{k,f}^{a,-}, e_{k,f}^{a,f}}^{\text{grd}} + y_{e_{k,f}^{a,f}, e_{k,f}^{a,+}}^{\text{grd}} = 0 \quad \forall k \in K, \forall f \in F_k^C \quad (4)$$

$$\sum_{s \in S_k} \chi_s^{\text{str}} y_s^{\text{str}} + \sum_{e \in E_k^G} \chi_e^{\text{grd}} y_e^{\text{grd}} \leq n_k \quad \forall k \in K \quad (5)$$

$$y_s^{\text{str}} \in \{0, 1\} \quad \forall s \in S \quad (6)$$

$$y_e^{\text{grd}} \geq 0 \quad \forall k \in K, \forall e \in E_k^G \quad (7)$$

The objective is to minimize the total costs of the selected flight strings (which include flight, fleet, and maintenance costs) plus the fleet costs of airplanes on ground at the count time. Additionally, the model contains flight cover constraints (2), departure and arrival flow balance constraints (3 and 4), and fleet size constraints (5).

A solution of *FARP-TS* given by y^{str} and y^{grd} defines a set of fleet-specific rotations, each comprising a set of flight strings and complies with all basic constraints FA-1 to FA-3 as well as RP-1 to RP-3. Specifically, the constraints FA-1-COVER and FA-2-FLEET are enforced by the model constraints 2 and 5, respectively. The definitions of the timeline network and flight strings take care of the fleet-flight compatibility constraint FA-3-COMP as well as the flight connection and ground time constraints RP-2-CONN and RP-3-GROUND. Since the network is cyclic and all nodes are transshipment nodes (observe that the right-hand side of the flow balance constraints 3 and 4 is 0), the flow y^{str} on a flight string arc is passed on from arc to arc until it finally arrives again at this arc, thereby enforcing constraint RP-1-CYCLE. As, by definition, each flight string is a maintenance-feasible sequence of flights, the solution complies with the maintenance regulations for A-checks, i.e. with the optional constraints RP-4-MNT and RP-5-MNTHUB.

In the sequel we base our integrated model on the flight string based model *FARP-TS*. Note that the associated flight based model, denoted as *FARP-T* is appropriate if maintenance constraints are not imposed. It can be considered as a special case of model *FARP-TS* by

- Introducing (non-augmented) flight strings of the size 1, i.e. flight strings which contain only a single flight, for each feasible combination of flight $f \in F^C$ and fleet $k \in K$,

- Neglecting maintenance costs in the calculation of the fixed string costs c_s^{str} ,
- Defining the event time of arrival events as the arrival time of the flight plus the minimal ground time for the fleet at the destination.

3.3 Modeling the cargo routing problem

In the cargo routing problem we have to determine which demand to fulfill using which itineraries, given a network of legs stemming from an operational schedule. This problem can be formulated as a multi-commodity flow problem with the set of O&Ds as commodities (cf. [Antes et al 1998](#)). There are two approaches. While in the arc-flow model decision variables represent flow on legs, in the path-flow model decision variables are introduced for all O&D-itineraries, i.e. sequences of legs which form a path connecting the origin with the destination. The path flow model has the advantage that several practical constraints on feasible itineraries can be controlled by defining the set of variables appropriately and hence can be handled outside the optimization model.

Let P be the set of all cargo flow paths/itineraries and P_l be the set of cargo flow paths that use leg l , i.e.

$$P_l := \left\{ p \in P \mid \exists k \in \{1, \dots, n_p\} : l_k^p = l \right\}.$$

Given the continuous decision variables x_p^{flow} which measure the amount of cargo transported over each path $p \in P$ in weight units, the path-flow based cargo routing model *CRP-P* is defined as follows.

Model CRP-P

$$\max \sum_{p \in P} m_p x_p^{\text{flow}} \quad (8)$$

$$\text{s.t.} \quad \sum_{p \in P_{\text{od}}} x_p^{\text{flow}} \leq d_{\text{od}} \quad \forall \text{od} \in OD \quad (9)$$

$$\sum_{p \in P_l} x_p^{\text{flow}} \leq w_l^{\max} \quad \forall l \in L \quad (10)$$

$$\sum_{p \in P_l} \text{vol}_{\text{od}_p} x_p^{\text{flow}} \leq v_l^{\max} \quad \forall l \in L \quad (11)$$

$$x_p^{\text{flow}} \geq 0 \quad \forall p \in P \quad (12)$$

The objective (8) is to maximize contribution to profit, and thus for every path the amount of flow is multiplied by the appropriate contribution margin. The demand constraints (9) restrict for all $\text{od} \in OD$ the amount which can be transported over all paths in P_{od} . Leg capacities are respected by constraints (10) and (11) for the weight and the volume capacity, respectively.

A solution of *CRP-P* complies with the basic constraints CR-1 to CR-6. Specifically, model constraints (9) cover the O&D demand constraints CR-1-DEMAND, and constraints (10) and (11) enforce the payload constraints CR-2-PAYLOAD. The remaining constraints CR-4 to CR-6 are assumed to be observed by each cargo flow path $p \in P$.

Also, path-flow models contain significantly less rows compared to arc-flow models while the number of path variables is usually significantly higher than the number of arc variables. Yet, one can observe that in an optimal solution, each O&D is usually transported on a single or two different cargo flow paths. Consequently, the number of cargo flow paths which are part of a solution is very small compared to the number of all feasible cargo flow paths, and thus model *CRP-P* can be solved using the well-known (delayed) column generation approach. Thus, the integrated models presented in the following chapter are based on model *CRP-P*.

3.4 Integrated models for the pragmatic planning approach

In this section, we present a model for the *ACSP* which is based on the models for fleet-ing/rotation planning and cargo routing presented in the previous sections. The model integrates the subproblems flight selection, fleet assignment, aircraft rotation planning, and cargo routing and it is adapted to support the pragmatic planning approach. For the separate models, we have assumed that the main input is a list of mandatory cargo flights in case of the *FARP* and a predefined schedule with O&D requests in case of the *CRP*. Several modifications are necessary: To support the pragmatic planning paradigm we differentiate between mandatory and optional cargo flights, i.e. only cargo flights $f \in F^C$ with $\text{mdt}_f = 1$ have to be covered, and cargo may only be routed on flight legs of selected flights. Also, different from iterative and separate modelling, the maximal payload weight and volume on a cargo flight leg are not known in advance since they depend on the fleet type the flight is assigned to. Finally, the objective is to maximize network-wide profit. Given all variables, sets, and parameters as defined for the models *FARP-TS* and *CRP-P*, we obtain the following integrated model *ACSP-T*:

Model ACSP-T

$$\begin{aligned} \max \quad & \sum_{p \in P} m_p x_p^{\text{flow}} - \sum_{s \in S} c_s^{\text{str}} y_s^{\text{str}} \\ & - c_k^{\text{acr}} \left(\sum_{s \in S} \chi_s^{\text{str}} y_s^{\text{str}} + \sum_{e \in E^G} \chi_e^{\text{grd}} y_e^{\text{grd}} \right) \end{aligned} \quad (13)$$

$$\text{s.t.} \quad \sum_{p \in P_{\text{od}}} x_p^{\text{flow}} \leq d_{\text{od}} \quad \forall \text{od} \in \text{OD} \quad (14)$$

$$\sum_{p \in P_l} x_p^{\text{flow}} \leq w_l^{\text{max}} \quad \forall l \in L^E \quad (15)$$

$$\sum_{p \in P_l} \text{vol}_p x_p^{\text{flow}} \leq v_l^{\text{max}} \quad \forall l \in L^E \quad (16)$$

$$\sum_{p \in P_l} x_p^{\text{flow}} - \sum_{s \in S_{f_l}} w_{k_s}^{\max} y_s^{\text{str}} \leq 0 \quad \forall l \in L^C \quad (17)$$

$$\sum_{p \in P_l} \text{vol}_{\text{od}_p} x_p^{\text{flow}} - \sum_{s \in S_{f_l}} v_{k_s}^{\max} y_s^{\text{str}} \leq 0 \quad \forall l \in L^C \quad (18)$$

$$\text{mdt}_f \leq \sum_{s \in S_f} y_s^{\text{str}} \leq 1 \quad \forall f \in F^C \quad (19)$$

$$\sum_{s \in S_{k,f}^+} y_s^{\text{str}} - y_{e_{k,f}^{\text{d},-}, e_{k,f}^{\text{d},f}}^{\text{grd}} + y_{e_{k,f}^{\text{d},f}, e_{k,f}^{\text{d},+}}^{\text{grd}} = 0 \quad \forall k \in K, \forall f \in F_k^C \quad (20)$$

$$- \sum_{s \in S_{k,f}^-} y_s^{\text{str}} - y_{e_{k,f}^{\text{a},-}, e_{k,f}^{\text{a},f}}^{\text{grd}} + y_{e_{k,f}^{\text{a},f}, e_{k,f}^{\text{a},+}}^{\text{grd}} = 0 \quad \forall k \in K, \forall f \in F_k^C \quad (21)$$

$$\sum_{s \in S_k} \chi_s^{\text{str}} y_s^{\text{str}} + \sum_{e \in E_k^G} \chi_e^{\text{grd}} y_e^{\text{grd}} \leq n_k \quad \forall k \in K \quad (22)$$

$$x_p^{\text{flow}} \geq 0 \quad \forall p \in P \quad (23)$$

$$y_s^{\text{str}} \in \{0, 1\} \quad \forall s \in S \quad (24)$$

$$y_e^{\text{grd}} \geq 0 \quad \forall k \in K, \forall e \in E_k^G \quad (25)$$

The objective (13) is to maximize the network-wide profit which is calculated as follows: for each cargo flow path $p \in P$, we multiply the flow x_p^{flow} on the path by the contribution margin m_p . Furthermore, we subtract the costs c_s^{str} of each selected flight string as well as the costs of airplanes on ground at the count time. Constraints (14) are identical to the O&D demand constraints (9) of model *CRP-P*. In contrast to model *CRP-P*, we now differentiate between maximal leg weight/volume constraints for external flight legs and for cargo flight legs. In the case of external flight legs $l \in L^E$, the maximal payload in terms of weight and volume is bounded by the maximal weight $w_l^{\max}$ and volume $v_l^{\max}$ of the external flight (constraints 15 and 16). In the case of cargo flight legs $l \in L^C$, constraints 17 and 18 link the continuous flow variables x^{flow} with the binary flight string variables y^{str} . They prohibit flow on any cargo flow path $p \in P_l$ using a leg l which belongs to a flight f_l that is not selected, i.e. a flight for which all string variables $y_s^{\text{str}}, s \in S_{f_l}$ which contain the flight take the value 0. If flight f_l is part of a flight string s with $y_s^{\text{str}} = 1$, the maximal payload is limited by the maximal weight $w_k^{\max}$ and volume $v_k^{\max}$ of the fleet type k_s . Flight covering constraints (19) require each mandatory cargo flight, i.e. every flight $f \in F^C$ for which $\text{mdt}_f = 1$ to be covered by a flight string while no flight may be covered more than once. Constraints (20) to (22) are identical to the departure and arrival flow balance constraints (3) and (4) as well as the fleet size constraints (5) of model *FARP-TS*.

Now, a solution of *ACSP-T* selects a set of optional cargo flights, assigns the mandatory and selected optional flights to fleets, generates rotations for each fleet, and identifies the optimal cargo flow paths for all O&D pairs. Following the reasoning given in the previous subsections, a solution is feasible with respect to all basic constraints FA-1 to FA-3, RP-1 to RP-3, and CR-1 to CR-6.

4 Model extensions

In this section, we shortly outline four model extensions which are relevant in practice. For a detailed discussion of their implementation and further model extensions, we refer to [Friedrichs \(2010\)](#).

4.1 Fixed costs for external flights

In the integrated models, we assume that fixed costs on external flights are not decision relevant, i.e. all external flights are considered as available for transportation. This is a reasonable assumption in many scenarios. For example, the passenger flight schedule of a mixed carrier is usually constructed independently of the cargo flight schedule. In other scenarios, the cargo capacity on an external flight operated by a partner i.e. a (cargo) airline or road-feeder services (RFS) has to be booked in advance for a fixed price. Including fixed costs for external flights into the model allows a cargo airline to evaluate which flights/destinations should be operated by its own fleet or by a partner airline/RFS which potential external capacities should be booked to transport cargo.

Let c_f^{flt} be the fixed costs of external capacities/flights $f \in F^E$. Then, we introduce additional binary decision variables $y_f^{\text{ext}} \in \{0, 1\}$ for external flights which take the value 1 if the external flight f is selected, and 0 otherwise. To consider the costs of external flights in the integrated models, we include the term

$$- \sum_{f \in F^E} c_f^{\text{flt}} y_f^{\text{ext}} \quad (26)$$

in the objective function. Additionally, the capacity constraints on legs of external flights $l \in L^E$ are modified in the following way:

$$\sum_{p \in P_l} f_p - w_l^{\max} y_{f_l}^{\text{ext}} \leq 0 \quad \forall l \in L^E \quad (27)$$

$$\sum_{p \in P_l} \text{vol}_p f_p - v_l^{\max} y_{f_l}^{\text{ext}} \leq 0 \quad \forall l \in L^E. \quad (28)$$

4.2 Cargo handling costs and constraints

Cargo handling costs may account for a significant part of the overall shipping costs. These costs are incurred when cargo is exported or imported at its origin and destination airports, or when cargo is transferred between two airplanes. Considering export and import costs and times at the origin and destination airports of an O&D request is straightforward and possible without any model modifications by adjusting the availability time and the maximal delivery time of od accordingly and including the additional import and export costs in the margin calculation of an O&D path. Yet, the costs of transferring (and storing, if necessary) cargo between flights as well as any time constraints for cargo transfers depend on the actual rotations of cargo airplanes and

external flights and hence in most cases it is not possible to decide a priori if a transfer (XFR) or a transit (TRA) of cargo is necessary between two flights. Consequently, additional variables and constraints have to be introduced into the model to account for transfer costs and to comply with time constraints. Here, we have to distinguish several cases of optional and mandatory transfers, respectively. The development of the additional constraints is quite involved and is outside the scope of this paper. For more details, we refer to [Friederichs \(2010\)](#).

4.3 Frequency constraints

The projected demand, i.e. the definition of the set of O&D requests, is often based on the assumption that the origin and destination airports of an O&D request are visited a minimal number of times during the planning period. The motivation for this is that an airline must offer a minimal service level to maintain its market position. Let $\epsilon_{s,a}$ be the number of times an airport a is visited in all flights belonging to a flight string $s \in S$. Then, introducing the *minimal frequency constraints*

$$\sum_{s \in S_a} \epsilon_{s,a} y_s^{\text{str}} \geq \text{fr}_a^{\min} \quad \forall a \in A \quad (29)$$

guarantees a solution in which each airport a is visited at least $\text{fr}_a^{\min}$ times.

4.4 Equal aircraft utilization

Some airlines try to ensure an equal *wear and tear* for all airplanes by requiring that every aircraft of a fleet operates all the flights assigned to its fleet, thus a single rotation is assigned to each fleet which is followed by all aircrafts of the fleet. These constraints can only be modeled in a connection network since the timeline model does not capture the flow of individual aircrafts. [Friederichs \(2010\)](#) has developed an integrated model ACSP-C based on the connection network. Here, equal aircraft usage constraints similar to the *subtour elimination constraints* of the Traveling Salesman Problem (TSP) (cf. [Dantzig et al 1954](#)) which enforce such a single rotation per fleet can be introduced. For more details see [Friederichs \(2010\)](#).

5 The branch and price and cut procedure at a glance

Large integer/mixed-integer programs such as our integrated model for the ACSP are commonly solved using a *branch and price and cut* approach. Branch and price and cut is a combination of the well-known methods branch and bound, column generation, and cutting planes. A major reason for using this procedure for solving ACSP is that the decomposition into a (*restricted*) *master problem* and *subproblems* for generating flight strings and cargo flow paths lends itself to the problem structure as schedule level constraints are handled in the master problem and constraints related to flight strings and cargo flow paths are handled in the subproblems. Although branch and price, i.e. the combination of branch and bound and column generation, has been

applied successfully to various integer problems such as vehicle routing (Desrosiers et al 1984; Chabrier 2006; Azi et al 2010) or crew scheduling (Freling et al 2004), Lübbecke and Desrosiers (2005) note that the implementation of such a procedure for a specific problem “is still an issue. This is not so much because of the complex interaction of components, but because of the vast possibilities to tune each of them”. Adding cutting planes during the solution procedure clearly leads to additional complexity since synthesizing the column and row generation process is a non-trivial task. Consequently, successful applications of the branch and price and cut procedure are still scarce. An introduction to branch and price and cut as well as findings of an application to multi-commodity flow problems are given by Barnhart et al (2000). Recently, Jepsen et al (2008) describe an application to vehicle routing.

To solve the integrated model ACSP-T, we designed and implemented a parallel branch and price and cut framework which allows to tackle problem instances of realistic sizes. The procedure is customizable, i.e. it can be used in various *exact* and *approximate* configurations, and adapts itself to the specific model and extensions considered in a particular planning situation.

Figure 3 depicts the flow with the main steps of the branch and price and cut procedure which we have developed. The procedure starts with a general initialization phase in which the problem data are preprocessed and several data structures used throughout the solution process as well as the restricted master problem *RMP* are constructed. *RMP* is a LP relaxation of model ACSP-T which contains only a subset of all flight strings S and cargo flow paths P . While P may initially be empty as demand must not be satisfied in our model, S must at least contain trivial flight strings, i.e. flight strings consisting of a single flight, for each mandatory flight to obtain an initial feasible solution and start the column generation procedure. In our approach, we keep a single instance of *RMP* in memory throughout the solution procedure and adapt it at each node by removing incompatible columns and rows rather than setting up a new *RMP* at each node. The main procedure starts with the selection of the next node.

At the beginning of each *node solution procedure*, the graphs for the pricing problems (G^P and G^S) have to be adapted to avoid generating columns which conflict with existing branching decisions. Note that instead of removing nodes and arcs in G^P and G^S , we introduce a boolean attribute $visible_v$ for all nodes v and $visible_e$ for all arcs e in both graphs which are initialized accordingly at each node. Thus, it is simple to restore the graphs at the beginning of each initialization procedure. After these initialization steps, *RMP* is solved. An integer *RMP* solution might still be infeasible with respect to certain constraints. In this case, we add any such violated constraints and resolve *RMP*. Constraints are added until the integer solution becomes feasible or the solution becomes fractional due to the new constraints. Note that we do not add columns and cuts in a single iteration but we alternate between pricing and cutting iterations. The reason for this is that the dual values associated with cuts influence the reduced costs of columns. Thus, if both columns and cuts are added in a single iteration, the new cuts may turn some of the identified improving columns non-improving, i.e. some of the columns may be generated and added needlessly.

In a single pricing iteration, we solve both the pricing problem for cargo flow paths and the pricing problem for flight strings in parallel. If improving cargo flow

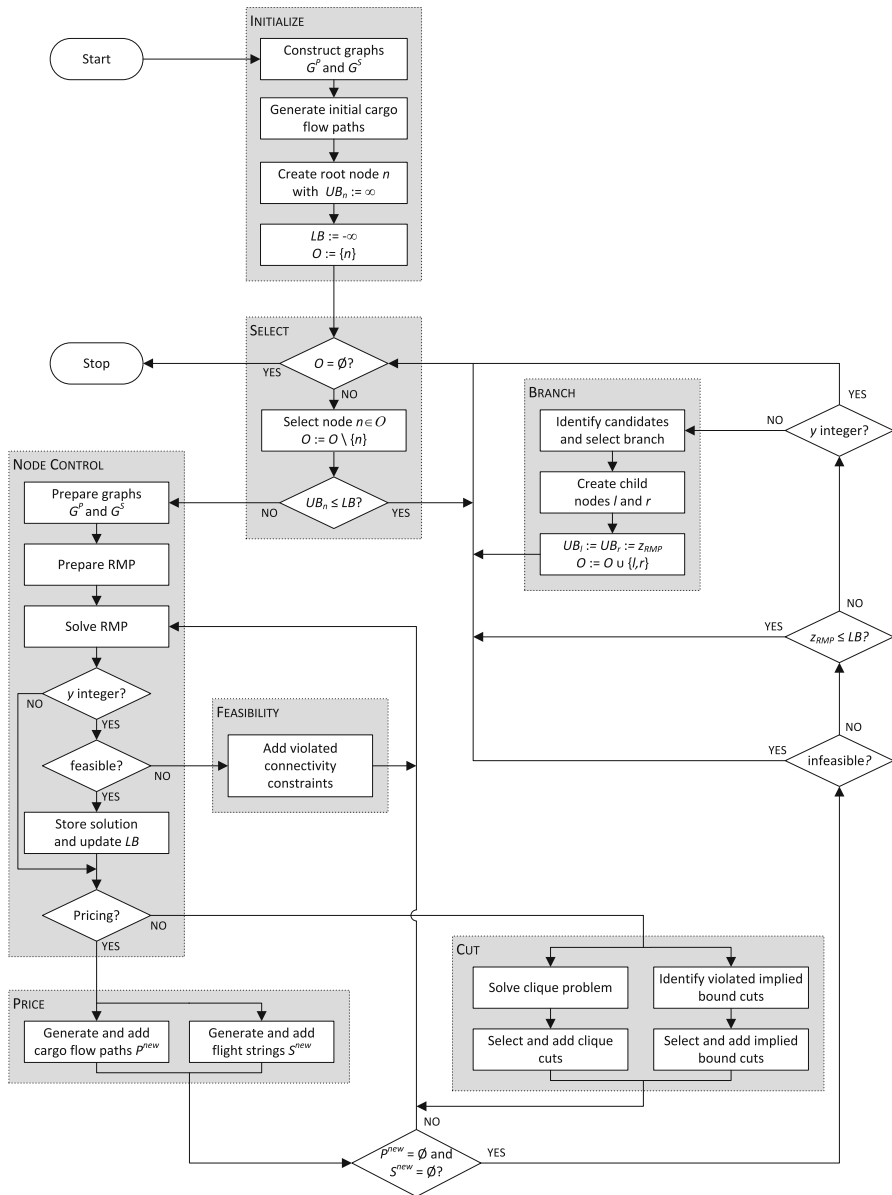


Fig. 3 The branch and price and cut procedure

paths P^{new} and/or improving flight strings S^{new} are found, we add P^{new} and S^{new} to RMP and continue with the next iteration. Otherwise, we stop the node solution procedure. Hence, the single termination criterion for the node solution procedure is that no improving columns are found. In a cutting iteration, we first identify all violated implied bound cuts and then select and add the most violated cuts. Regardless

of whether or not cuts have been added, the node solution procedure continues with a pricing iteration.

Once the node solution procedure stops, i.e. once $P^{\text{new}} = \emptyset$ and $S^{\text{new}} = \emptyset$, we first check if the node can be pruned. If the node is pruned, we continue with the node selection step. Otherwise, we identify all possible branching candidates and select the most promising candidate according to the branching strategy (see Sect. 7).

In the following section, we describe the main modules of the branch and price and cut procedure, i.e. the pricing, the branching, and the cutting modules, as well as the most important acceleration/approximation techniques in greater detail. We refer to Sect. 9 for the results obtained using such improvement techniques.

6 Column generation: solving the pricing problem

During the column generation phase, we have to check for improving columns of two different types: cargo flow path columns and flight string columns. Thus we have to solve two different pricing problems, *PRICE-P* to find improving cargo flow path columns, and *PRICE-S* to find improving flight strings. Note that the optimal solution to an LP-relaxation is found only if both pricing procedures stop with the testimonial for no improvement.

6.1 Solving the cargo flow pricing problem PRICE-P

The purpose of the cargo flow pricing problem is to find columns/variables x_p^{flow} with positive reduced costs. Given the dual value π_{od_p} associated with the O&D demand constraint of a path p as well the dual values π_l^w and π_l^v associated with the maximal weight and maximal volume constraints of all legs in p , the reduced costs $\bar{c}_p$ of p are defined as

$$\bar{c}_p = r_{\text{od}_p} - \pi_{\text{od}_p} - \sum_{l \in L_p} (c_l^{\text{leg}} + \pi_l^w + \text{vol}_{\text{od}_p} \pi_l^v). \quad (30)$$

Thus, the reduced costs $\bar{c}_p$ of a path p are positive if the revenue r_{od_p} of the associated O&D od_p outweighs all cargo handling and transportation dependent costs and dual values. All dual values involved in the calculation of $\bar{c}_p$ are positive because they are associated with less-equal-constraints only which restrict the flow on p . Note, that the calculation of $\bar{c}_p$ becomes more complex if additional constraints (e.g. hub transfer constraints) are inserted into the model.

Now, a solution of RMP is optimal for the embedded cargo routing problem if no path p with $\bar{c}_p > 0$ exists (remember that the solution may still be suboptimal as there might still be improving flight strings not introduced into RMP yet). Note that for a given $\text{od} \in \text{OD}$, finding the path with maximal reduced costs is equal to finding the path with minimal variable transportation and handling costs plus dual values.

We model the pricing problem as a (resource constrained) shortest path problem on a directed graph $G^P = (V^P, E^P)$ with V^P as the set of all legs L and E^P representing

feasible connections between legs. A connection from a leg $l_1 \in L$ to a *successor leg* or *successor* $l_2 \in L$ is feasible if

- (a) both legs are consecutive legs of the same flight f_{l_1} or
- (b) both legs belong to different flights ($f_{l_1} \neq f_{l_2}$), the destination airport of l_1 is the origin airport of l_2 ($\text{dest}_{l_1} = \text{orig}_{l_2}$), and the destination airport is a transfer airport ($\text{hub}_{\text{dest}_{l_1}} = 1$).

Given the graph G^P and an O&D od , we have to find the shortest path from all legs l_1 which leave from the origin airport of od ($\text{orig}_{l_1} = \text{orig}_{od}$) and all legs l_2 which arrive at the destination airport of od ($\text{dest}_{l_2} = \text{dest}_{od}$).

Due to the periodic nature of a flight schedule, the graph may contain cycles, i.e. arcs which connect to a leg node earlier in time. Routing cargo from a leg to another leg earlier in time implies that the cargo is delivered in the subsequent planning period. Note that although the graph is cyclic, we can use Dijkstra's (1959) algorithm to find a shortest path since all weights/costs on arcs are non-negative, i.e. the cost function is monotonically increasing with each path extension.

Yet, a (shortest) path may not be feasible since the maximal delivery time constraint of the O&D od_p may be violated. The delivery time can be interpreted as a limited *resource* whose consumption increases with each path extension. Thus, we have to solve a *resource-constrained shortest path problem* (RCSPP), which is NP-complete even for the case of a single resource (Handler and Zang 1980). The standard solution algorithm for RCSPP is still based on Dijkstra's shortest path algorithm and employs a labeling technique to handle resource constraints. Additionally, we apply *dominance tests* to reduce the number of labels. For details on the method we refer to Desrochers and Soumis (1988).

To accelerate the path generation of the labeling algorithm we have implemented the A*-strategy introduced by Hart et al (1968). The idea is to use problem-specific heuristic information on the length/cost of path extensions. For any path from the origin airport o to a labeled node v in the graph we distinguish between two cost components,

- The costs of the partial path from o to v denoted $g(v)$, and
- A heuristic estimate of the costs of the shortest path from v to the destination d denoted $h(v)$.

Here the heuristic value $h(v)$ must be *admissible*, i.e. it must not overestimate the length of the shortest path from v to d . Now, the algorithm works analogously to Dijkstra's algorithm, with the only difference that we always extend the partial path from the labeled node v with minimal value for $g(v) + h(v)$. In fact, Dijkstra's algorithm can be seen as a special case of A* with $h(v) = 0$.

Obviously, the better the lower bound given by $h(v)$ is, the faster the algorithm will find the shortest path. An intuitive way to calculate a rather rough lower bound is to use the *great-circle distance*¹ between airports multiplied by some predefined transportation cost per km. In our approach, we calculate more realistic estimates $h(v)$ using the length of the shortest (not necessarily resource constrained) path from v to d .

¹ The *great-circle distance* is the shortest distance between any two points on the surface of a sphere.

This estimate is precalculated during the initialization phase (INITIALIZE) for all pairs of nodes by solving an unconstrained all-pairs shortest path problem with respect to variable transportation costs on the connection network G^P .

In a similar way, it is possible to improve the running time of the algorithm further by calculating lower bounds for the time needed to transport goods from v to d to enable to detect paths from o to v which cannot be extended to feasible od-paths. To compute these lower bounds, we again solve an all-pairs shortest path problem during the initialization phase, but this time with the duration of legs (dur_l) and the connection times between legs ($\text{time}(\text{arr}_{l_1}, \text{dep}_{l_2})$) as costs on leg nodes and connection arcs.

6.2 Solving the flight string pricing problem PRICE-S

The purpose of the pricing problem for flight strings is to find columns/variables y_s^{str} with positive reduced costs. Given the dual value π_{k_s} associated with the maximal number of aircraft constraints of fleet k_s as well as the dual values $\pi_{k_s, f_1^s}^{\text{dep}}$ and $\pi_{k_s, f_{n_s}^s}^{\text{arr}}$ associated with the departure and arrival flow balance constraints of the first and last flight in s , the reduced cost $\bar{c}_s$ of a flight string $s \in S$ can be written as

$$\begin{aligned} \bar{c}_s = & -\chi_s^{\text{str}}(c_{k_s}^{\text{acr}} + \pi_{k_s}) - \sum_{f \in F_s} c_{k_s, f}^{\text{flt}} + \sum_{l \in L_s} (w_{k_s}^{\text{max}} \pi_l^w + v_{k_s}^{\text{max}} \pi_l^v) \\ & - \sum_{f \in F_s} \pi_f - \pi_{k_s, f_1^s}^{\text{dep}} + \pi_{k_s, f_{n_s}^s}^{\text{arr}}. \end{aligned} \quad (31)$$

Note that some of the terms involved in equation (31) are positive (e.g. $\sum_{l \in L_s} (w_{k_s}^{\text{max}} \pi_l^w + v_{k_s}^{\text{max}} \pi_l^v)$), some are negative (e.g. $-\sum_{f \in F_s} c_f$), and some may be positive or negative (e.g. $-\sum_{f \in F_s} \pi_f$). As in the case of cargo flow paths, the calculation of $\bar{c}_s$ has to be adapted if additional constraints are added to the model.

Now, a solution of *RMP* is optimal for the embedded fleet assignment and aircraft rotation problem if no flight string s with $\bar{c}_s > 0$ exists. In the following, we use the term *PRICE-S* to refer to the problem of identifying improving flight strings. Again, we model the pricing problem as a constrained shortest path problem. However, in contrast to *PRICE-P*, we use a directed acyclic graph $G^S = (V^S, E^S)$ to cope with the problem of negative edge weights due to negative dual values. In general, G^S has to represent n^{mnt} weeks, where n^{mnt} is the maximal number of weeks a single flight string can touch. We calculate n^{mnt} using the maximal elapsed time since the last maintenance stop, which is given by $\text{mnt}_k^{\text{elapsed}}$, as follows:

$$n^{\text{mnt}} := \left\lceil \frac{\max_{k \in K} \text{mnt}_k^{\text{elapsed}}}{N} \right\rceil.$$

The node set V^S consists of three types of nodes:

- fleet nodes v_k^{source} for each fleet type $k \in K$ which serve as source nodes,
- flight nodes v_f^i for each cargo flight $f \in F^C$ and each week $1 \leq i \leq n^{\text{mnt}}$, and

- a *sink node* v^{sink} representing the mandatory maintenance stop after the last flight in a flight string.

For each flight node v , we define f_v as the flight represented by v . The set of arcs E^S represents all feasible connections between flights. For each fleet node v_k^{source} , we include arcs to all flight nodes v_f^1 of the first week whose associated flight f departs from an airport orig_f which is a maintenance station for the fleet type ($\text{hub}_{k, \text{orig}_f}^{\text{mnt}} = 1$). Let gt_a be the minimal ground time of all fleets k at airport a , then, we include arcs from each flight node v_f^i of week i to

- All flight nodes $v_{f'}^i$ of the same week i if $\text{dest}_f = \text{orig}_{f'}$ and $\text{arr}_f + \text{gt}_{\text{arr}_f} \leq \text{dep}_{f'}$,
- All flight nodes $v_{f'}^{i+1}$ of the following week $i+1$ if $\text{dest}_f = \text{orig}_{f'}$, $\text{arr}_f + \text{gt}_{\text{arr}_f} \leq N + \text{dep}_{f'}$ and $f \neq f'$, and
- The sink node v^{sink} .

Note that we do not include connections from a week i to a week j with $j > (i+1)$ since such connections imply an unproductive ground time of more than a week, and consequently the same schedule can be carried out with one airplane less. Given a connection from a node v to a node v' , we call v *predecessor node* and v' *successor node*.

With G^S defined as above, PRICE-S can be formulated as a resource constrained shortest path problem. Similar to the situation in PRICE-P, a flight string may not be feasible since maintenance constraints may be violated. Thus the constraints on the elapsed time *eltime*, the flight time *ftime*, and the number of cycles *cycles* define the resources limiting path construction. Additionally, each path must be *elementary*, i.e. it may only visit each flight $f \in F$ once.

We use the standard label-pulling algorithm for directed acyclic graphs. The algorithm draws on the property that it is possible to define a topological order among all nodes v according to the *node time* t_v which is defined as follows. For flight nodes v of week i , t_v is set to $\text{dep}_{f_v} + (i-1) \cdot N$. For source nodes v^{source} and the sink node v^{sink} , t_v is set to high negative values and a high positive value, respectively. Note that the order among source nodes is not important since no source node is connected to any other source node. In the beginning, an initial label is created and added to the source node v_k^{source} of each fleet k . Then, the algorithm visits each node v in the topological ordering and pulls all predecessor labels whose extension to the current node is feasible. For example, the flight represented by the current node must be compatible with the current aircraft type. Since the topological ordering among nodes guarantees that all predecessor nodes have been visited, it is sufficient to visit each node exactly once. Note that in this basic version, the algorithm identifies *all* improving paths in each iteration, which is computationally intractable even for small problem instance. To be able to solve real-size problem instances, we apply the following relaxation and acceleration techniques.

The number of labels generated at each node can be enormous since all labels of all predecessor nodes must be extended to a successor node. To reduce the number of improving flight strings being generated in a single pricing iteration, we only

keep the best $\sigma_{\text{MaxLabels}}^S$ labels at each node v (*label limitation*). When limiting the number of labels, *label ordering* is used to propagate the best labels only. Here, a common approach is to order the labels at a node lexicographically, i.e. first according to (reduced) costs, and then according to resource consumption which in our case means to consider the consumptions of the resources *fltime*, *eltime*, and *cycles* as the second, third, and fourth ordering criterion, respectively. During our computational tests, we found that most of the flight strings generated in a single iteration are almost identical except for one or two flights. Grönkvist (2005) makes similar findings considering an aircraft routing problem. This has led to the concept of bucket ordering to diversify the search in the subproblem and to generate columns which are unlike each other. Here we first categorize the labels at a node v into buckets according to their predecessor flight and then propagate only the label with highest reduced cost from each bucket. Note that an extension would be to propagate more than one label from each bucket.

Obviously, limiting the number of labels may lead to suboptimal solutions. Hence, $\sigma_{\text{MaxLabels}}^S$ must be set to a reasonably high value to be able to obtain good solutions. If it is set too low, the algorithm might fail to find paths with positive reduced costs since all partial paths which potentially lead to improving flight strings might be discarded at an early node v because other partial paths which later turn out to be infeasible or non-improving have lower costs at v . Additionally we apply (heuristic) *dominance tests* considering the costs, the elapsed time, the flight time, and the number of cycles to discard supposedly non-efficient labels.

7 Branching strategies

The reference model ACSP-T contains binary flight string variables y_s^{str} for each flight string $s \in S$. In an optimal solution of *RMP*, some of these variables may attain fractional values. In the classical branching approach one would branch directly on such a variable by setting the variable to 0 or to 1 in the child nodes. We refer to this as *flight string branching*. Due to the fact that flight strings usually span several weeks (depending on the maintenance constraints), a solution only consists of a few flight strings. Consequently, branching on flight strings results in a highly unbalanced branch and bound tree. The 1-branch reduces the solution space considerably while the impact of the 0-branch is almost negligible in view of the enormous number of feasible flight strings. However, with a depth-first node selection strategy which follows the 1-branch first, feasible integer solutions are found relatively fast.

Another, problem-specific branching approach is related to the different ASCP-subproblems to be solved. The fundamental goal of the ACSP is to design an optimal flight schedule by identifying the optimal combination of optional flights. Now, an optional flight f may not be selected entirely in an optimal solution of *RMP*. *Optional flight branching* forces or forbids f to be selected in the child nodes. Let dist_l be the distance (in km) of a leg l , dist_p be the total distance (in km) of a cargo flow path p , and $\text{load}(f)$ be the maximal payload on any leg of f according to the current solution.

Then, we calculate a score for each optional flight as follows:

$$\text{score}(f) := \sum_{i=1}^{n_f} \sum_{p \in P_{l_i^f}} m_p \cdot f_p \cdot (\text{dist}_{l_i^f} / \text{dist}_p) - \min_{k \in K : \text{comp}_{k,f}=1 \wedge w_k^{\max} \geq \text{load}(f)} c_{k,f}^{\text{flt}}$$

The first term of the right-hand side estimates the cumulated margin (i.e. revenue variable transportation costs) on all legs l of flight f . We approximate this margin by considering only the fraction $\text{dist}_{l_i^f} / \text{dist}_p$ of the margin of a cargo flow path p using leg l which can be attributed to l . The second term considers the fixed costs of the flight. As the assignment of flights to fleets is not known yet, we select the fleet with minimal fixed costs $c_{k,f}^{\text{flt}}$ which has a capacity equal to or higher than the maximal payload on any legs of f . Thus, the score indicates the profitability of a flight. Now, we select the most profitable optional flight $f \in F^C$ with a fractional value to branch on and adapt the corresponding flight covering constraint accordingly.

The second problem-specific branching type is related to the fleet assignment problem. In a fractional solution of RMP, a cargo flight $f \in F^C$ may be assigned to several flight strings of different fleet types. *Fleet-flight branching* forces f to be covered by a specific fleet $k \in K$ in the 1-branch, and forbids f to be covered by fleet k in the 0-branch. Let $S_{k,f} = \{s \in S_f | k_s = k\}$ be the set of those flight strings conducted by fleet k which cover cargo flight f . Given a fractional solution of RMP, we calculate the score for each fleet-flight pair as $\text{score}(k, f) := \sum_{s \in S_{k,f}} y_s^{\text{str}}$. Then, we select the fleet-flight pair (k, f) with the highest fractional score and force f to be covered by k in the 1-branch.

The last problem-specific branching type is a variant of the Ryan-Foster branching for set partitioning problems (c.f. [Ryan and Foster 1981](#)) which is commonly called *follow-on branching*. Given the current solution of RMP and $S_{i,j}$, i.e. the set of those flight strings which cover cargo flight $j \in F^C$ immediately after $i \in F^C$, we calculate for each pair of cargo flights $i, j \in F^C$ the follow-on-score $\text{score}(i, j) := \sum_{s \in S_{i,j}} y_s^{\text{str}}$ and we select the flight pair (i, j) with the highest fractional score. Then, we require in the 1-branch that any flight string which covers flight i must cover flight j immediately after i while in the 0-branch no flight string may cover flights i and j consecutively. Note that at the time we start to branch on follow-ons, each (optional) cargo flight takes a value of 0 or 1. Hence, a flight pair (i, j) with a fractional score implies that there must be flight strings which cover i or j but not both.

Note that when using the model extension for external flights, it may still be necessary to branch on external flight variables. Since external flights are represented by single variables, we simply apply traditional branching in this case.

8 Cutting planes

Adding cutting planes during the solution procedure has two effects which are desirable in the context of branch and bound. First, they improve the bound obtained by the LP relaxation, and second, they push the solution of the LP relaxation towards integrality. First computational tests showed that the LP-optimum at the root of the

branch and bound tree has a high number of fractional flight strings and the gap to the first integer solution found may be up to 40 %. Also the upper bounds obtained during the solution procedure are rather poor primarily because they do not accurately represent the fixed costs of optional cargo flights. The average load factor on cargo flights within the industry is about 60–65 %. Since an optional flight does not need to be covered in a solution of *RMP* (as long as the flight is not set to 1 due to a branching decision), the flight may take a fractional value large enough to transport the maximal payload of all legs of the flight, resulting in unaccounted fixed costs. Now the concept of *implied bound cuts* which has been introduced by Hoffman and Padberg (1991) has shown to improve the accounting of fixed flight costs. In the following we describe the adaption of this concept to the ASCP. Specifically, we show how to identify violated inequalities and discuss their impact on the pricing problems.

As an example, consider a solution of *RMP* with an O&D pair $od \in OD$ whose demand weight of 5 tonnes is shipped via a single path $p_1 \in P$, i.e. $x_{p_1}^{\text{flow}} = 5$. Now, assume that p_1 and several other paths $p_2, \dots, p_n$ use a leg l of an optional cargo flight $f \in F^C$, and the only flight string variable y_s^{str} which covers flight f takes the value 0.5. Furthermore, suppose that the associated fleet type k_s has a maximal payload weight of 100 tons. Then, the O&D demand constraint for od reads as follows

$$x_{p_1}^{\text{flow}} \leq 5 \quad (32)$$

and the maximal leg weight constraint for l has the following form

$$x_{p_1}^{\text{flow}} + x_{p_2}^{\text{flow}} + \dots \leq 100 \cdot y_s^{\text{str}}. \quad (33)$$

Since y_s^{str} must take a value of 0 or 1 in any integer feasible solution and $x_{p_1}^{\text{flow}}$ is bounded by 5 the following inequality

$$x_{p_1}^{\text{flow}} \leq 5 \cdot y_s^{\text{str}}. \quad (34)$$

is valid as it holds for every feasible integer solution and it cuts off the current solution $x_{p_1}^{\text{flow}} = 5$, $y_s^{\text{str}} = 0.5$, i.e. it is a violated valid inequality.

In general, an implied bound cut is associated with an O&D $od \in OD$ and an optional cargo flight $f \in F^C$. Note that the combined flow of all cargo flow paths $p \in P_{od}$ which use any leg of f is bounded by d_{od} . Let $P_{od,f}$ be the set of cargo flow paths for O&D od which use flight f , i.e.

$$P_{od,f} := \left\{ p \in P_{od} \mid \exists k \in \{1, \dots, n_p\} : f_{l_k^p} = f \right\}.$$

Then, the implied bound cut for O&D od and flight f takes the form

$$\sum_{p \in P_{od,f}} x_p^{\text{flow}} \leq \sum_{s \in S_f} y_s^{\text{str}} \cdot d_{od} \quad (35)$$

Since a single O&D od may be transported via several cargo flow paths, the cumulated flow on paths $p \in P_{od,f}$ which use an optional flight f may be lower than the

demand d_{od} . Consequently, $\sum_{s \in S_f} y_s^{\text{str}}$ must not attain the value 1 to satisfy the corresponding implied bound cut. However, as will be shown in our computational results, implied bound cuts have a tremendous impact on the bound obtained and accelerate the overall solution procedure significantly.

Due to the fact that the overall number of potential implied bound cuts is huge, we do not add implied bound cuts a priori but identify and add violated implied bound cuts during each cutting iteration. The identification of violated implied bound cuts is straightforward. The separation algorithm loops through every combination of an O&D od and an optional flight f and calculates the left-hand side as well as the right-hand side of Eq. (35). If the left-hand side is bigger than the right-hand side, a cutting plane has been identified and is added to the list of all violated cuts. Having identified all violated cuts, we add the n^{ibc} most violated cuts per iteration.

Given an optimal primal and dual solution of the RMP with cut inequalities, the dual values associated with implied bound cuts must be included in the calculation of the reduced costs $\bar{c}_p$ of cargo flow paths (Eq. 30) and the reduced costs $\bar{c}_s$ of flight strings (Eq. 31). Let $\pi_{od,f}^{\text{ibc}}$ be the dual value associated with the implied bound cut for an O&D od and an optional cargo flight f . Given the set of optional cargo flights F_p used by a cargo flow path p , i.e. $F_p := \{f \in F^C \mid \text{mdt}_f = 1 \wedge \exists i \in \{1, \dots, n_p\} : f_{l_i^p} = f\}$, we include the term

$$- \sum_{f \in F_p} \pi_{od,f}^{\text{ibc}}$$

in the formula for the reduced cost $\bar{c}_p$ of a cargo flow path p as well as the term

$$+ \sum_{f \in F^C : \text{mdt}_f = 1} \sum_{od \in OD} \pi_{od,f}^{\text{ibc}} \cdot d_{od}$$

in the formula for the reduced $\bar{c}_s$ of a flight string s . Then, the modifications of the shortest path procedures for solving the pricing problems are straightforward.

9 Computational experiments

In this section, we present the computational results obtained with our approach. For that purpose we have created a set of realistic problem instances representing different planning scenarios in different types of cargo airlines which altogether mimic the potential applications of our approach in real-world air cargo scheduling. We first describe the different airline types which have been classified from the set of air cargo carriers. Then we shortly describe the different scenarios resulting in a basis of 60 problem instances for our computational study before we finally report and evaluate the obtained results.

9.1 Classification of airline types

Cargo airlines vary with respect to the markets served and the kind of cargo transported. Some cargo airlines specialize in frequent short-haul connections, other airlines operate long-haul flights to transport bulk cargo internationally, and still other airlines focus on express deliveries using a distributed hub network. Naturally, these differences are reflected in the network structure and the flight schedule of a cargo airline. To evaluate the performance of our model and solution procedure for a wide range of cargo airlines, we define four *airlines* with each one representing a different type of carrier with different scope and freight characteristic and thus serving as a substitute for a major real-world airline from the respective segment. Note that the models and solution procedures have been developed for typical cargo airlines such as Cargolux, Lufthansa Cargo, or Singapore Airlines Cargo which do not specialize in express services and operate a fleet of 10–25 cargo aircrafts. To be able to evaluate how the approach behaves for instances of express carriers, we also generated instances for an express carrier such as DHL or Federal Express. We drew upon general data such as reports on the air cargo industry as well as airline specific information such as timetables and annual reports to generate networks, demand data, and production costs and mimic a specific type of cargo airline. Table 1 gives an overview of our four fictitious airlines. In the following, we describe each of the airlines.

9.1.1 RC: a regional cargo carrier

The first airline represents a regional carrier such as Air Hong Kong or Tampa Cargo. The airline mainly operates direct round trips, i.e. flights which start at the hub in Hong Kong (HKG), fly to the destination airport where cargo is picked up and/or delivered, and return to the hub. Since the flights are relatively short and cargo can usually be delivered using a sequence of two legs with a single stop at the hub, the airline is able to guarantee short delivery times.

9.1.2 IC: an international cargo carrier

The second scenario reflects the situation of a major international cargo-only airline such as Cargolux. In contrast to the regional carrier, the flights which start and end at the airline's hub in Luxembourg (LUX) are rather long and visit several destinations. For example, a typical route could start at LUX and visit the destinations Barcelona (BCL), Mexico City (MEX), Houston (IAH), and New York (JFK) before returning to LUX. In addition to its own fleet, the airline utilizes road-feeder services (RFS) to collect cargo at several smaller destinations and bring it to one of its main airports which are frequently visited by cargo flights.

9.1.3 MI: a mixed carrier

The third scenario represents a typical mixed carrier such as Lufthansa or Singapore Airlines with a separate cargo division. Hence, airline MI maintains a passenger fleet as well as a dedicated cargo fleet. MI may route cargo to destinations not served by its

Table 1 Airline scenarios

Airline		RC	IC	MI	EX
Characteristics					
Type		Cargo only	Cargo only	Mixed	Cargo only
Scope		Regional	Intl.	Intl.	Intl.
Main freight		Cargo	Cargo	Cargo	Express
Delivery time (days)		1–3	2–7	2–7	1–3
En-route stops		1–2	2–5	2–5	1–2
External flights		–	RFS	PAX	–
Network					
Destinations	Cargo	23	69	42	142
	External	–	77 (RFS)	81 (PAX)	–
	<i>Total</i>	23	134	93	142
Hubs		HKG	LUX	SIN, BRU	BRU, BAH, PHD, PTY
Freighter	B747	–	16	13	14
	B757	–	–	10	31
	B727	–	–	–	11
	A300	9	–	–	28
	<i>Total</i>	9	16	23	84
Typical schedule					
Cargo flights (avg.)	90	80	120	800	
Revenue TKM	13,788,512	90,962,599	111,849,662	119,806,624	
Offered TKM	20,749,359	136,775,422	173,540,852	191,806,624	

Typical schedule information (including revenue and offered tonne-kilometers (TKM)) are weekly

own fleet using the bellyhold capacity of passenger flights (PAX). The airline operates a main hub in Singapore (SIN) and a smaller hub in Brussels (BRU) which is used as a gateway to markets in North and South America as well as to consolidate cargo streams from America/Europe to and from Asia.

9.1.4 EX: an express carrier

In the last scenario, we consider the network structure of an international express carrier such as FedEx, UPS, or DHL. In our case, the airline operates a network of four hubs in each of its core regions. These hubs are located in BRU for the region Europe/Africa, Bahrain (BAH) for the region Middle East/Asia, Philadelphia (PHD) for the region North America, and Panama City (PTY) for the region South America. The flights of airline EX are similar to that of airline RC in that they usually consist of two or three legs which connect a hub with one or two destinations. However, since EX operates internationally, the number of flights offered weekly and the total number of flights and airplanes is significantly higher compared to airline RC.

9.2 Planning scenarios and data sets

For each of the airlines described above, we defined three *planning scenarios* leading to three *data sets* of problem instances. The data sets of an airline vary with respect to the number of mandatory and optional cargo flights and the number of external passenger/road-feeder services flights. We generated five problem instances for each scenario resulting in 60 problem instances in total. Based on the information on the network and flight structure as well as demand structure of the associated real airline realistic O&D's and flights have been generated randomly. Table 2 gives an overview of the generated data sets. In general, the number of potential combinations of optional flights rises with each set. In the following, we briefly describe the scenarios and data sets, respectively.

- Set I The instances of sets I represent a planning scenario close to operations in which the number of mandatory cargo flights is relatively high. Hence, only a few of the optional cargo flights will be part of a schedule. These instances are typical for planning situations where an existing flight schedule is adapted to reflect small demand changes.
- Set II The instances of sets II are based on the instances of set I, with the only difference that the number of optional flights was doubled. Note that set II instances extend the set I instances, i.e. we only generated the additional optional flights.
- Set III Set III instances differ from set II instances with respect to the ratio of mandatory to optional flights and represent primarily strategic planning scenarios. The flights are the same as in the set II instances, yet half of the mandatory flights are now marked as optional.

Table 2 Data sets

Set	Cargo flights			Ext. flights	O&Ds	Legs (avg.)
	Mandatory	Optional	Sum			
RC-I	70	30	100	—	800	247
RC-II	70	60	130	—	800	321
RC-III	35	95	130	—	800	321
IC-I	60	30	90	220	3,500	779
IC-II	60	60	120	220	3,500	893
IC-III	30	90	120	220	3,500	893
MI-I	70	30	100	300	4,000	1,123
MI-II	70	60	130	300	4,000	1,217
MI-III	35	95	130	300	4,000	1,217
EX-I	720	180	900	—	6,000	2,046
EX-II	720	360	1,080	—	6,000	2,469
EX-III	360	720	1,080	—	6,000	2,469

Each set consists of five instances. The total number of legs is averaged as flight routes and hence the number of legs vary by instance

9.3 Computational results

All algorithms have been implemented using the Microsoft .NET framework, version 3.5, with the C# programming language. IBM ILOG CPLEX 12.1/single-core version is used to solve the linear programming relaxations. All tests were carried out on a quad-core Intel Xeon Processor (E5430) with 2.66 GHz and 14 GB of RAM. However, to demonstrate that the model and solution procedure can be used on standard workstation hardware with up to 4 GB main memory (RAM), we did not apply branch and bound parallelization since parallel branch and bound requires significantly more main memory. As we generated five instances per data set, we report only average values for all instances of a set. For the instances of some sets—especially for instances in set III which are characterized by a high number of optional flights and a small number of mandatory flights—it takes a long time to prove the optimality of a solution. Thus, we restrict the running time of tests without maintenance constraints to 4 h. With maintenance constraints, we set a limit of 8 h for instances of airlines RC and IC and a longer limit of 12 and 16 h for the instances of airlines MI and EX, respectively.

Table 3 summarizes the results of solving model *ACSP-T* with and without maintenance constraints. In the first part, we show the results obtained for *ACSP-T* without maintenance constraints and using the CPLEX MIP solver with all cargo flow paths pre-generated. In the second part, we present the results without maintenance constraints using our exact branch and price and cut procedure. The results of solving *ACSP-T* with maintenance constraints using the approximative version of our branch and price and cut procedure are reported in the last part. When solving *PRICE-S*, we limit the number of labels per node to 20 and use the bucked ordering strategy with a single label per predecessor node as we obtained the best results using these settings. We aborted the solution procedure as soon as a gap of 0.5 % was reached. The table contains information on the running time, the objective value, and the gap of the first and best integer solution found. The overall running time is given in the last column.

The CPLEX MIP solver was only able to solve the small problem instances of airline RC. For airlines IC, MI, and EX, the number of feasible cargo flow paths is too high, and hence CPLEX aborted with a memory error. For the small instances of airline RC, CPLEX is considerably faster than our exact branch and price and cut procedure. This is not surprising as no costly column generation iterations are performed during the solution process. For example, set RC-II is solved to optimality in 12 s using CPLEX, while our procedure requires 81 s. Instances of set RC-III are solved to optimality in 2,463 s using CPLEX. Contrarily, the gap using our solution procedure still amounts to 1.61 % after the solution procedure was terminated.

Using our exact approach, we are able to find high quality integer solution for the instances of airlines IC, MI, and EX in short times. In many cases, solutions with an optimality gap of less than 1 % are found within 20 min. In the worst case, the first integer solution with a gap of 0.76 % is found after 4,879 s or 81 min (set MI-III). The average gap of the first integer solution amounts to 0.71 %, which is subsequently reduced to 0.33 %. We mainly attribute these excellent results in terms of solution quality to the application of implied bound cuts.

Table 3 Final results using model ACSP-T with and without maintenance constraints

Set	First integer			Best integer			Time (total)
	Time	OV	Gap (%)	Time	OV	Gap (%)	
ACSP-T, all cargo flow paths pre-generated							
RC-I	—	—	—	—	1,999,935	0.00	1
RC-II	—	—	—	—	2,221,666	0.00	12
RC-III	—	—	—	—	2,810,027	0.00	2,463
ACSP-T, exact branch and price and cut							
RC-I	3	1,999,440	0.34	4	1,999,935	0.00	6
RC-II	10	2,215,482	1.25	41	2,221,666	0.00	81
RC-III	40	2,790,618	2.16	2,641	2,805,937	1.61	14,401
IC-I	53	16,366,110	0.45	53	16,366,110	0.00	163
IC-II	165	17,999,598	0.74	181	18,009,648	0.00	780
IC-III	499	19,547,858	1.42	3,335	19,635,847	0.85	11,796
MI-I	332	23,407,428	0.45	1,288	23,438,513	0.06	5,646
MI-II	1,275	23,915,834	0.38	4,759	23,935,746	0.28	14,427
MI-III	4,879	25,283,001	0.76	9,227	25,306,497	0.66	14,486
EX-I	291	28,201,301	0.06	3,971	28,205,578	0.01	4,569
EX-II	688	34,755,241	0.20	7,991	34,768,099	0.17	14,408
EX-III	2,965	42,211,058	0.34	5,238	42,212,030	0.34	14,403
ACSP-T+MC, approximate branch and price and cut							
RC-I	453	1,986,440	0.59	9,805	1,989,440	0.45	11,822
RC-II	1,562	2,199,170	1.65	8,124	2,206,202	1.34	23,405
RC-III	3,899	2,763,981	2.69	13,786	2,770,981	2.48	28,835
IC-I	1,393	16,175,117	0.72	3,003	16,195,117	0.61	12,431
IC-II	10,839	17,417,787	3.02	22,437	17,543,934	2.42	29,766
IC-III	15,975	19,249,979	2.15	24,251	19,299,979	1.94	28,898
MI-I	11,192	23,160,514	0.59	14,329	23,163,597	0.55	25,845
MI-II	23,736	23,587,531	0.42	24,032	23,602,816	0.39	32,155
MI-III	28,540	24,624,619	4.92	35,517	24,625,871	4.55	43,225

When considering maintenance constraints we are not able to find feasible integer solutions for airline EX within the maximal running time of 16 h with the approximative version of branch and price and cut. Even an increase to 32 h did not lead to any improvement. For these instances, it takes more than 3 h to price out the root node, and the running times of the RMP rise dramatically to several minutes per iteration. Since airline EX operates 84 aircrafts and more than 700 flights per week, significantly more branching decisions must be made to find the first integer solution. For all instances of the other airlines, maintenance feasible integer solutions could be found within the running time limits. However, the running times to the first integer solutions are on average 65 times higher compared to the case that no maintenance constraints are considered. In the best case,

the first integer solution is found after 453 s (set RC-I), while it takes more than 28,000 s in the case of set MI-III. The average gap of the first integer solutions to ASCP-T with maintenance constraints is 1.86 %, which is subsequently reduced to 1.71 %. Thus, we can say that good-quality solutions are obtained within 12 h for all instances.

As can be seen by comparing the results of sets I with those of sets II and III, the impact of the number of optional flights and the impact of the ratio of optional flights to mandatory flights on the running times and the solution quality is significant. For ASCP-T without maintenance constraints, the running times to the first integer solution increase by 207 % on average from sets I to sets II, and by 283 % from sets II to sets III. However, the average gap only increases from 0.02 % for sets I to 0.11 % for sets II and 0.86 % for sets III. Similar effects are obtained for ACSP-T with maintenance constraints.

Our final conclusions are that our integrated model and the solution procedures are applicable at real-world cargo airlines. Even with complicating constraints such as maintenance constraints, the problem instances of major cargo airlines can be solved within reasonable time. The limitations of a model with maintenance constraints are revealed when solving the instances of airline EX which represents a huge express carrier and could not be solved within our time limits. As mentioned before, these instances were only generated for evaluation purposes, while our research focus lies on typical cargo carriers such as Lufthansa Cargo or Singapore Airlines Cargo.

The evaluations of the various algorithmic features show that the orchestration of the complex solution procedure is a difficult task. However, we could identify a dominant setting in most cases. In the following, we report on the impact of the main algorithmic features which result in significant improvements of the solution quality and/or running time.

9.3.1 Implied bound cuts

Implied bound cuts have a considerable impact on the upper bound which was improved by 3.17 % on average and the objective value of best integer solution found which increased by 1.76 % on average. Consequently, the gap between the best bound and the best integer solution found could be reduced dramatically from 5.70 % on average without implied bound cuts to only 0.33 % with cuts. Note that the gap between the first integer solution and the best bound is up to 40 % for some problem instances when solved without cuts. As implied bound cuts can only be applied to optional flights, a problem instance that contains more optional flights and less mandatory flights results in a higher impact of implied bound cuts. This can be observed when comparing the values of sets I, II, and III. While the average gap of the instances of sets I, i.e. the sets with a high number of mandatory flights, amounted to 1.63 % without implied bound cuts, the average gap of the instances of sets III with a high number of optional flights significantly rose to 10.84 %. With implied bound cuts, this value could be reduced to 0.86 %. We attribute the benefits of implied bound cuts to the fact that they specifically address the interdependencies between network structure decisions (i.e. flight selection) and flow decisions (i.e. cargo routing).

9.3.2 Branching strategies

The average gap of the first integer solution found is 19.43 % when using the traditional branching strategy, compared to 1.37 % when using the problem-specific strategy. However, following the problem-specific branching strategy, the running time to the first integer solutions rises considerably (e.g. from 3,431 s to 11,183 s in case of set IC-II) as the branching decisions only have a small impact on the solution space and hence more branching decisions must be made to find the first solution. Yet, the average gap of the best integer solution found is still significantly lower (1.19 %) compared to the gap of 7.67 % when using the traditional branching strategy.

9.3.3 PRICE-S

When limiting the number of labels, the bucket ordering strategy consistently leads to better results compared to canonical ordering. Even with a limitation to 20 labels, the results are impressive compared to canonical ordering with 100 labels. While the average objective value increases slightly by 0.5 % on average, the average time required to price out the root node falls sharply by more than 50 %. When canonical ordering is used with 20 labels, no feasible solution could be found for many problem instances. For the same reason, dominance tests in *PRICE-S* prove to be indispensable. For more details on the effect of different numbers of improving flight strings added, different values for label limitation, as well as ordering strategies we refer to [Friedrichs \(2010\)](#).

9.3.4 PRICE-P

Using A* instead of Dijkstra's algorithm reduces the running time of *PRICE-P* dramatically. On average, only 58 % of the time compared to Dijkstra's algorithm is needed.

9.3.5 Subproblem synchronization

Due to varying running times of *PRICE-S* and *PRICE-P* for specific problem instances, the node running time can be reduced significantly if *PRICE-S* (*PRICE-P*) is not solved in every pricing iteration. For example, the node solution time for instances of airline IC could be reduced by more than 50 % when solving *PRICE-S* only every fifth column generation iteration. On the other hand, in case of airline RC it is best to solve *PRICE-S* in each iteration.

9.3.6 Parallelization

The potential benefits of a parallel branch-and-bound implementation are enormous as the productive CPU time spend to solve nodes scales almost linear with the number of workers. Concerning the gap obtained within a specified time, moving from a single thread to 2 threads leads to a 50 % reduction on average, and a further increase to 4 threads brings an additional reduction of 25 % on average.

10 Final remarks

In our paper, we have extensively studied the complex decision problems arising at cargo airlines when planning the flight schedule for the next period of operation. A solution to this problem determines the transportation network as well as the resource consumption. From a marketing point of view, it limits the revenue potential, and from a production point of view, it fixes processes and a large fraction of operational costs. The complexity of the problem stems from the fact that several subproblems which are of combinatorial and computational complexity in itself have to be addressed. The approach which we propagate is a pragmatic one since it assumes that in order to maintain continuity in operations and service, planning is not carried out from scratch on the level of legs. Rather, the idea is that analyzing the existing schedule results in the specification of flights which remain or become mandatory and flights which are new options or become optional. Consequently, the combinatorial problem is to determine the optimal combination of mandatory and optional flights. For this optimization problem we have developed and presented an integrated model which covers the subproblems of schedule design, fleet assignment, rotation planning, and cargo routing. In current airline planning practice and in most approaches presented in literature, these subproblems are treated sequentially and iteratively. Our approach leads to mixed-integer programs which in realistic scenarios are of extremely high dimensionality. We have developed and presented a state-of-the-art branch and price and cut procedure which is shown to be effective and efficient for problem instances of realistic structure and size. We have also shown how complex constraints on maintenance and handling costs can be adequately modeled and fulfilled. The result of our research has been built into the model/method component of a fully productive Decision Support System. With regard to future research directions, we believe that integrating further schedule design aspects into our model may have a positive influence on the generated schedule. For example, timing decisions could be modeled explicitly (rather than implicitly by adding additional optional flights) by associating departure time windows with each flight, similar to the approach presented by [Desaulniers et al. \(1997\)](#).

Acknowledgments We would like to thank two anonymous referees for their constructive criticism and valuable comments.

References

- Antes J (1998) Modellbasierte Analyse von Frachtflugplänen—Konzeption, Implementierung und Anwendungspotentiale eines Bewertungsmodells für Frachtflugpläne. PhD thesis, Department of Information Systems and Operations Research (WINFORS), University of Cologne, Cologne
- Antes J, Campen L, Derigs U, Titze C, Woll GD (1998) SYNOPSIS: a model-based decision support system for the evaluation of flight schedules for cargo airlines. *Decis Support Syst* 22(4):307–324
- Azi N, Gendreau M, Potvin JY (2010) An exact algorithm for a vehicle routing problem with time windows and multiple use of vehicles. *Eur J Oper Res* 202(3):756–763
- Barnhart C, Cohn A (2004) Airline schedule planning: accomplishments and opportunities. *Manuf Service Oper Manag* 6(1):3–22
- Barnhart C, Boland NL, Clarke LW, Johnson EL, Nemhauser GL, Shenot RG (1998) Flight string models for aircraft fleet assignment and routing. *Transp Sci* 32(3):208–220
- Barnhart C, Hane CA, Vance PH (2000) Using branch-and-price-and-cut to solve origin-destination integer multicommodity flow problems. *Oper Res* 48(2):318

- Barnhart C, Belobaba P, Odoni AR (2003) Applications of operations research in the air transport industry. *Transp Sci* 37(4):368–391
- Chabrier A (2006) Vehicle routing problem with elementary shortest path based column generation. *Comput Oper Res* 33(10):2972–2990
- Dantzig G, Fulkerson R, Johnson S (1954) Solution of a large-scale traveling-salesman problem. *Oper Res* 2(4):393–410
- Desaulniers G, Desrosiers J, Dumas Y, Solomon MM, Soumis F (1997) Daily aircraft routing and scheduling. *Manag Sci* 43(6):841–855
- Desrochers M, Soumis F (1988) A generalized permanent labeling algorithm for the shortest path problem with time windows. *INFOR* 26(3):191–212
- Desrosiers J, Soumis F, Desrochers M (1984) Routing with time windows by column generation. *Networks* 14(4):545–565
- Dijkstra EW (1959) A note on two problems in connexion with graphs. *Numerische Mathematik* 1(1): 269–271
- Etschmaier MM, Mathaisel DFX (1985) Airline scheduling: an overview. *Transp Sci* 19(2):127–138
- Freling R, Lentink RM, Wagelmans AP (2004) A decision support system for crew planning in passenger transportation using a flexible branch-and-price algorithm. *Ann Oper Res* 127:203–222
- Friedrichs S (2010) Air Cargo Schedule Planning. PhD thesis, Department of Information Systems and Operations Research (WINFORS). University of Cologne, Cologne
- Gopalakrishnan B, Johnson EL (2005) Airline crew scheduling: state-of-the-art. *Ann Oper Res* 140: 305–337
- Gopalan R, Talluri KT (1998) Mathematical models in airline schedule planning: a survey. *Ann Oper Res* 76(1–4):155–185
- Grönkvist M (2005) The tail assignment problem. PhD thesis, Department of Computer Science and Engineering, Chalmers University of Technology and Göteborg University, Göteborg
- Handler GY, Zang I (1980) A dual algorithm for the constrained shortest path problem. *Networks* 10(4):293–309
- Hart PE, Nilsson NJ, Raphael B (1968) A formal basis for the heuristic determination of minimum cost paths. *IEEE Trans Syst Sci Cybern* 4(2):100–107
- Hoffman KL, Padberg M (1991) Improving LP-representations of zero-one linear programs for branch and cut. *ORSA J Comput* 3(2):121
- Jepsen M, Petersen B, Spoorendonk S, Pisinger D (2008) Subset-row inequalities applied to the vehicle-routing problem with time windows. *Oper Res* 56(2):497–511
- Li D, Huang HC, Morton A, Chew EP (2006) Simultaneous fleet assignment and cargo routing using benders decomposition. *OR Spectr* 28(3):319–335
- Lohatepanont M, Barnhart C (2004) Airline schedule planning: integrated models and algorithms for schedule design and fleet assignment. *Transp Sci* 38(1):19–32
- Lübbecke ME, Desrosiers J (2005) Selected topics in column generation. *Oper Res* 53(6):1007–1023
- Marsten RE, Muller MR (1980) A mixed-integer programming approach to air cargo fleet planning. *Manag Sci* 26(11):1096–1107
- Ryan DM, Foster BA (1981) An integer programming approach to scheduling. In: Wren A (ed) *Computer scheduling of public transport: urban passenger vehicle and crew scheduling.*, North-Holland, Amsterdam, pp 269–280
- Tang CH, Yan S, Chen YH (2008) An integrated model and solution algorithms for passenger, cargo, and combi flight scheduling. *Transp Res Part E: Logist Transp Rev* 44(6):1004–1024
- Yan S, Chen SC, Chen CH (2006) Air cargo fleet routing and timetable setting with multiple on-time demands. *Transp Res Part E: Logist Transp Rev* 42(5):409–430
- Zils M (2000) Modellbasierte und prozessorientierte Flugplanung. PhD thesis, Department of Information Systems and Operations Research (WINFORS). University of Cologne, Cologne